MATH INQUIRERS HOLIDAY PROGRAM

Math Inquirers P3 Tutor Guide

Holiday program for the Primary 3 level

Notice it. Test it. Explain why.

PRIMARY 3 HOLIDAY PROGRAM

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Program guide for tutors: schedule, lesson routine, ten minute-by-minute lesson scripts with answer keys, a progress tracker and a parent summary. Math Inquirers is online first: every lesson runs live on video; the face-to-face notes are kept for partner centres.

The program at a glance

Level	Children whose school year covers Primary 3 topics (ages 9 to 10), going into P4, who are ready for more thinking than standard practice
Length	10 lessons of 90 minutes, two a week over the 5 weeks of the holidays
Group size	Up to 8 children per tutor, live online. Pilot with 6. Add a second tutor or helper above 8.
Focus	Thinking skills: making a list in order, guess and check, drawing a model, working backwards, finding patterns, reasoning and explaining. Not speed drills.
The arc of every lesson	Start concrete with objects, record in a table, spot the pattern, state the rule, use it (often backwards), then explain why.
What children take home	A student workbook with an inquiry journal, 10 worksheets and a certificate
Before tutors start	Every tutor signs the tutor confidentiality agreement before receiving any materials.

Schedule

Week	Lesson	Title	Topic	Big idea
Week 1a	1	**Number Sleuths**	Numbers to 10 000	Place value to 10 000
Week 1b	2	**Cryptarithm Crackers**	Addition and Subtraction	Column addition and subtraction with regrouping

Week 2a	3	**Times Table Treasure**	Tables of 6, 7, 8 and 9	Look for a pattern
Week 2b	4	**Model Makers**	Word Problems: Bar Models	Draw a model
Week 3a	5	**Remainder Riddles**	Division	Division with remainders
Week 3b	6	**Fraction Walls**	Fractions	Drawing a model (fraction wall
Week 4a	7	**Area Architects**	Area and Perimeter	Systematic listing
Week 4b	8	**Angle and Line Detectives**	Angles, Perpendicular and Parallel Lines	Organised counting
Week 5a	9	**Measure and Money Masters**	Length, mass, volume, money and time	Converting units
Week 5b	10	**Challenge Day**	Mixed puzzles: bar graphs and logic	Choosing a strategy

Suggested pairing: two lessons a week, for example Tuesday and Thursday, or Saturday and Sunday. Keep the same time slot all five weeks.

The 90-minute lesson routine

Every lesson uses the same routine so children know what to expect and tutors can prepare quickly.

Minutes	Part	What happens	Face to face	Online
0–10	**Opening and warm-up**	0 to 2 opening. 2 to 5 sneak-peek puzzles (own paper, nobody marked). 5 to 8 a quick game. 8 to 10 five quick questions.	Sit in a circle or horseshoe.	Cameras on. Sneak-peek answers written on own paper and shown to the tutor on camera in the open call (or kept to share if the child wants). Game answers by thumbs, hands or whiteboards: show your tutor on camera or keep it to share if you want.
10–25	**Explore**	Hands-on in pairs. Objects first, no writing yet.	Materials box on each table.	Each child has their own materials at home. Pairs go to breakout rooms.
25–35	**Mini-lesson**	Tutor builds the table on the board, names the rule and asks what stays the same and what changes.	Whiteboard.	Shared slide and annotation tool.
35–40	**Brain break**	Stand, stretch, drink water.	A short movement game.	Stand up on camera. Count the stretches together.
40–72	**Worksheet**	Page 1 (Start here) in pairs, short teaching, page 2 alone. The tutor visits every breakout room by minute 47. After page 1 and Q4 and Q5 of page 2 any child may choose Think harder, Go further or the Investigation (own slot, minutes 67 to 72). Tutor asks “how do you know?” rather than giving answers.	Printed workbook, tutor circulates.	Printed workbook. Child shows the page to the camera in the open call. The call is not private, so no private messages and no photos. Tutor checks one or two answers each.
72–80	**Talk time**	First 3 minutes: Investigation debrief (volunteers share what they noticed). Then share one answer and explain it. Compare methods.	Volunteers share. Passing is always fine.	Volunteers unmute. The tutor asks for volunteers first, never calls a child by name cold, and keeps no public tallies or rankings.

80–90	**Exit ticket, look back and stamp**	80 to 86 exit ticket on a slip of paper. 86 to 88 look back (2 minutes, short and safe, pass allowed). 88 to 90 stamp the inquiry journal and close.	Paper slip, tutor stamps.	Child writes the answer and shows it to the tutor on camera in the open call, or keeps it to share if they want. Tutor stamps the next time or posts it.

Teaching habits that matter

- Ask “How do you know?” after every answer, right or wrong. Never mark or compare children in front of the group.
- Wait 10 seconds before helping. Let children struggle a little.
- Praise the thinking (“You checked it two ways”), not just the answer.
- Let children disagree with each other and settle it with objects or a drawing.
- If a child wants more, point to Think harder, Go further or the Investigation, not extra drill. Never label children.

Sneak-peek puzzles, choice and look-back (every lesson)

- Sneak-peek puzzles (minutes 2 to 5): two small puzzles on ideas we have not learnt yet. Puzzle 1 is an access puzzle that any child can try with objects, a drawing or counting, so everyone has a first win. Puzzle 2 is a thinking puzzle. Say: “These are ideas we have not learnt yet. Guessing and I do not know are welcome. Nobody is marked.” Children write answers on their own paper and show them to you on camera in the open call, or keep them to share if they want (or all hold paper to the camera at the same moment, so nobody is singled out); never read out. The call is not private: no private messages, no photos.
- Choice, not sorting: there are no public labels. Every child gets the same Start here pages. After page 1 and Q4 and Q5 of page 2 any child may choose Think harder or Go further. The Investigation is open to everyone; its easy first step is printed on page 2 of Start here and on the Investigation page. You invite the WHOLE group, never one child: “When you have finished Start here you can choose Think harder, Go further or the Investigation.” You never announce who is on which page, and pairs are not split unless both children agree.
- Go further on the explore slide: a second what-if question is on the explore slide from minute 10 so that nobody waits. Any pair may take it up when their first task is done.
- Talk time: ask for volunteers first. A pass is always fine, and so is sharing with a partner. Do not cold-call by name. Do not keep public tallies or rankings.
- Look back (minutes 86 to 88, two minutes): Something I did well, The hardest part was, What helped me. A pass is always allowed. The page is never collected or read out; children keep it. You cannot see the paper, so tell the WHOLE group: “If you ticked still stuck, say check-in please in the open class, or ask your grown-up to tell me.” You then talk it through with that child in the open session, with a parent nearby. No hand signals that single a child out, never a private message.

Preparing the materials

- Send the workbook PDF to parents one week before. Ask them to print it single-sided and the cut-out pages on card if they can.

- Cut-outs used: digit cards (Lessons 1, 2), fraction wall (Lesson 6), square grid (Lesson 7), right-angle tester (Lesson 8), play coins and notes (Lesson 9).
- Keep the answer key with the tutor only. Check answers together on screen.
- Use the shared screen or a digital whiteboard for the mini-lesson table.

Setting up online

- Send parents the workbook and materials list one week before. Needed: the printed workbook and cut-outs, scissors, a ruler, paper and a pencil.
- Use one approved platform with breakout rooms and screen share. Admit only registered children and never share the link publicly.
- Ask the parent to sit nearby but not give answers. The camera should show the child's page and hands.
- Hold up the answer page to the camera at the end of each lesson, in the open group call. Never ask a child to send a photo or a message.
- Do not record lessons (this is in the tutor agreement). Add a second tutor or helper above 8 children.

Materials master list

Lesson	Title	Face to face	Online set-up
1	Number Sleuths	Digit cards 0–9 (two sets per child), a place value mat (Th H T O), mini whiteboards, a 10-sided die or a bag of digit cards 0–9 for the warm-up game. For the sneak-peek puzzles: paper, a pencil and a few counters or small objects for each child.	Children cut out the digit cards at the back of the workbook (or write 0–9 on small paper squares) and draw a Th H T O mat on paper. The tutor draws the digits on camera.
2	Cryptarithm Crackers	Digit cards 0–9 (one set per child), mini whiteboards, a printed 3 × 3 grid, counters or small paper squares to move around. For the sneak-peek puzzles: paper, a pencil and a few counters or small objects for each child.	Children use the cut-out digit cards at the back of the workbook (or paper squares numbered 0–9) and draw the grids on paper. Share the screen for the column sums and write on them.
3	Times Table Treasure	Multiplication square (1 to 10), dot paper, coloured pencils, a ruler, mini whiteboards. For the sneak-peek puzzles: paper, a pencil and a few counters or small objects for each child.	Tutor draws on a shared digit-circle slide. Each child draws 10 dots round a circle (like a clock) labelled 0 to 9, and has dot paper and coloured pencils.
4	Model Makers	Multilink cubes (about 30 per pair), strips of squared paper, scissors, mini whiteboards and markers. For the sneak-peek puzzles: paper, a pencil and a few counters or small objects for each child.	Tutor builds the bars on a shared slide. Each child uses Lego bricks, coins or paper strips at home, and a whiteboard or paper to draw bars. Print the workbook before the lesson. Pairs work in one breakout room for 10 minutes, then return for the check slide.
5	Remainder Riddles	25 counters per child (or beans or buttons), 6 small cups for groups, mini whiteboards and markers. For the sneak-peek puzzles: paper, a pencil and a few counters or small objects for each child.	Children use 25 small things at home (dried beans, buttons, LEGO bricks) and a few cups or bowls. Share the screen for the slides; children hold whiteboards up to the camera. Print the workbook before the lesson. Pairs work in one breakout room for 10 minutes, then return for the check slide.

6	Fraction Walls	The printed fraction strip page (7 strips per child, all the same length, 24 cm), coloured pencils, scissors, mini whiteboards. For the sneak-peek puzzles: paper, a pencil and a few counters or small objects for each child.	Send the printed fraction strip page instead of asking children to measure and cut strips. Marks every 12, 8, 6, 4, 3 and 2 cm give halves, thirds, quarters, sixths, eighths and twelfths. Share the screen for the slides. Print the workbook before the lesson. Pairs work in one breakout room for 10 minutes, then return for the check slide.
7	Area Architects	12 square tiles (or 12 squares cut from 1 cm grid paper) per child, 1 cm grid paper, mini whiteboards, a ruler. For the sneak-peek puzzles: paper, a pencil and a few counters or small objects for each child.	Children cut 12 squares from squared paper (or draw on squared paper) and hold up their rectangles to the camera; the tutor shares a grid slide to draw on.
8	Angle and Line Detectives	Scrap paper to fold a right-angle checker, a ruler, mini whiteboards, square grid paper. For the sneak-peek puzzles: paper, a pencil and a few counters or small objects for each child.	Each child folds a right-angle checker on camera and goes on a 2-minute right-angle hunt at home; the tutor shares the grid slides to draw on. Print the workbook before the lesson. Pairs work in one breakout room for 10 minutes, then return for the check slide.
9	Measure and Money Masters	Party menu cards (one per pair), play money, a measuring tape or metre ruler, a kitchen scale and a measuring jug if available, a clock with movable hands, mini whiteboards. For the sneak-peek puzzles: paper, a pencil and a few counters or small objects for each child.	Show the menu card on screen; children use paper and pencil (play money optional). Pairs work in breakout rooms and type their best party plan in the chat.
10	Challenge Day	Puzzle cards for 2 stations (list, measure), digit cards 0 to 9 (or paper cards), squared paper, mini whiteboards, certificates, stamps. For the sneak-peek puzzles: paper, a pencil and a few counters or small objects for each child.	Two stations, 7 minutes each. Pairs stay in one breakout room, and the tutor shares the next station card when the timer ends. No physical materials beyond paper. Logic grids and graphs are shared on screen and annotated together.

How to use the scripts

Every lesson below is a minute-by-minute script. The slides match it: one slide or two for each part. You do not have to say the words exactly, but keep the questions, because the questions are what makes children think.

SAY	Words to say aloud, in plain language for 8- and 9-year-olds. Said in your own voice, not read from the page.
DO	What you do: set up, hand out, show a slide, or move around the room.
ASK / EXPECT	Questions to put to the group, with the answer you are listening for. Let children answer before you explain.
IF STUCK	A ladder of help. Give hint 1, wait ten seconds, then give hint 2. Do not give the answer.

Five habits that run through every script

- Predict first. Before children test anything, ask for a guess and write it down. A wrong guess is useful because we can test it.
- Wait ten seconds after every question. Count silently. Most children answer in the last four.
- Ask “How do you know?” after right answers too, so that children do not read your face for the answer.
- Praise the thinking, not the answer: “You checked it two ways” beats “Well done”.
- When an answer is wrong, say “Let us test that,” and use objects, not “No”.

Answering a wrong answer

If...	Then...
A child gives a wrong answer	“Thank you. Let us test it with the counters.” Then build it together.
A child gives a right answer but cannot explain	“Show me with your counters.” Then: “Tell your partner how you know.”
A child says “I do not know”	“What do you notice? What could you try first?” Offer hint 1.
Two children disagree	“Both of you, convince the group using objects.” Let the objects decide.

Lesson 1: Number Sleuths

Week 1 (first lesson) • Numbers to 10 000 • Workbook: Lesson 1

Objective	Read and build 4-digit numbers by place value; make the greatest, smallest, odd, even and closest numbers from digit cards; list every number in order; solve clue puzzles and continue number patterns.
Thinking skills	Place value to 10 000, renaming (15 hundreds = 1 thousand 5 hundreds), making a list in order, looking for a pattern, using clues one at a time, explaining "I have them all"
Materials	Digit cards 0–9 (two sets per child), a place value mat (Th H T O), mini whiteboards, a 10-sided die or a bag of digit cards 0–9 for the warm-up game. For the sneak-peek puzzles: paper, a pencil and a few counters or small objects for each child.
Online set-up	Children cut out the digit cards at the back of the workbook (or write 0–9 on small paper squares) and draw a Th H T O mat on paper. The tutor draws the digits on camera.
Prepare	Solve the worksheet yourself first. Set out one materials box per pair. Open the slides for this lesson and check the answers in the speaker notes.

0–2 min Opening

SAY *"Welcome, Number Sleuths! Every number hides clues in its digits. Today we use the clues to find mystery numbers, build the biggest and the smallest, and prove that we have found every number."*

DO Greet each child by name. Show the lesson title slide. Read "Today we will…" aloud.

2–10 min Sneak-peek puzzles (2–5), warm-up game (5–8) and quick questions (8–10)

SAY *"Here are two sneak-peek puzzles. These are ideas we have not learnt yet, so guessing is welcome and so is "I do not know". Nobody is marked. Write your answer on your own paper. When I say "show me", hold it up to the camera for me."*

DO Show the "Sneak-peek puzzles" slide (minutes 2 to 5). Read puzzle 1 aloud and give about 1 minute: it is an access puzzle, so any child can try it with objects, a drawing or counting. Then puzzle 2, about 1 minute, the thinking puzzle. Look at the papers on camera quietly. Do not show the answers on the slide and do not read answers out. Say "thank you" to everyone in the same way.

Nobody is labelled. Never say who got it, who is on which page, or "fast group" or "top group". Do not single anyone out. Invite the whole group once Start here is done. It is an invitation, never a ranking: every child can choose Think harder or the Investigation once page 1 and Q4 and Q5 of page 2 are done. We come back to puzzle 2 in talk time, for volunteers.

Sneak-peek puzzle	Answer (tutor only) and what to listen for

Puzzle 1 (access): Take the digit cards 4, 0 and 9. Make the biggest 3-digit number you can. Move the cards around to try.	940. Listen for: Biggest digit first, then the next biggest: 9, 4, 0.
Puzzle 2 (thinking): Now use the cards 4, 0, 9 and 3, each once. What is the smallest ODD 4-digit number? Say how you know.	3049. Listen for: Odd means the ones digit is 9 or 3. 0 cannot go first. Try 3049 and 4093: 3049 is smaller.

Choice during the lesson (groups of 8, online)

- Minutes 10 to 40 (explore, mini-lesson, brain break): everyone does the same. The explore slide has a Go further what-if from minute 10, so a pair that has finished its first task has real work at once. Do not say who is ready for it.
- Minute 40: everyone opens Start here page 1 and works in pairs. A line at the top says: “Do page 1, then Q4 and Q5 on page 2. After that you may choose: Think harder, Go further or the Investigation.” You never announce who is on which page. Invite the whole group, never one child: “When you have finished Start here you can choose Think harder, Go further or the Investigation.” Do not split a pair unless both children agree.
- The first-step help for the Investigation is printed on page 2 of Start here, so every child sees it. Pairs work in breakout rooms, so visit EVERY room by minute 47 (about a minute each, camera on, a parent nearby) and ask “how do you know?”, not “is it done?”. At about minute 52 check in again with two or three children, including children on Start here.
- Minute 67: everyone chooses what to do next: finish page 2, Think harder, Go further or the Investigation. The Investigation has its own slot, minutes 67 to 72, and its debrief opens talk time. Everyone is back together for talk time at minute 72. An Investigation that is not finished goes home; volunteers share one thing at the start of next lesson.
- Next lesson everything starts again from the same pages: nobody is on a track.

DO Warm-up game (minutes 5 to 8). Now the game, then the quick questions (minutes 8 to 10): Biggest number wins. Draw four digits one at a time; children place each in Th, H, T or O straight away and cannot move it. Biggest number scores. Then play for the smallest.

SAY *“Draw a Th H T O mat. I will call out four digits, one at a time. Write each digit in a box straight away. No moving!”*

“Who made the biggest number? Where did you put the 9? Why there?”

DO After the game, ask the quick questions on the warm-up slides. Use thumbs or a whisper to partners before taking answers.

Ask	Expect
What is the value of the 7 in 4719?	700
3000 + 40 + 5 = ?	3045
What is 1 more than 4999?	5000

What is 100 less than 6050?	5950
Which is greater: 7089 or 7098?	7098

10–25 min Explore (in pairs, with objects)

SAY *"With the cards 2, 4, 6 and 8, how many 4-digit numbers between 4000 and 6000 can you make? Write a guess first."*

DO 1. Take the cards 2, 4, 6 and 8.

2. Make 4-digit numbers between 4000 and 6000.

3. Write each one on your whiteboard.

4. Which one is closest to 5000?

SAY *"Talk to your partner while you work. I will come round and ask you questions."*

IF STUCK Stuck: "Which card must go in the thousands place if the number is between 4000 and 6000?"

Missing some: "Keep the 4 in front. Now change the hundreds digit in order: 2, then 6, then 8."

Ready for more: the Go further what-if on the slide (cards 2, 4, 5, 8: 12 numbers between 4000 and 6000). Or: "How many numbers can you make altogether with the four cards?" (24: six start with each card.)

DO Pairs work from minute 10. If a pair finishes the first task, they may take the Go further what-if on the slide (do not say who). At about minute 15 bring the group back together and take the three questions below, one at a time (minutes 15 to 25).

Ask	Expect
How many numbers did you make?	6: 4268, 4286, 4628, 4682, 4826, 4862.
How do you know you have them all?	Every one must start with 4. Then list in order: hundreds digit 2, 6, 8, with two numbers each.
Which is closest to 5000?	4862, only 138 away. The nearest number above 5000 is 6248, which is 1248 away.

25–35 min Mini-lesson (you teach, they join in)

DO Show the mini-lesson slides. Think aloud as you work. Say what you notice, what you test and how you check. Children copy how you think.

SAY *"Greatest number from 4, 0, 8, 1. The thousands place is worth the most, so the biggest digit goes there: 8. Then 4, then 1, then 0: 8410."*

"Smallest. I want the smallest digit first, but 0 cannot go first or it is not a 4-digit number. So 1 goes first, then 0, then 4, then 8: 1048."

"Greatest odd number from 2, 9, 6, 0. An odd number ends in an odd digit. The only odd card is 9, so I put 9 in the ones place FIRST. Then the biggest of the rest, in order: 6, 2, 0. So 6209."

"3-digit numbers from 0, 3, 6. I fix the first digit: 3. Then 306, 360. Next first digit: 6. Then 603, 630. 0 cannot go first. So there are 4 numbers, and my list shows I have them all."

Worked example on the slides	Answer
Greatest and smallest with 4, 0, 8, 1. Greatest: biggest digit in the thousands place, then the next biggest. Smallest: the smallest digit that is not 0 goes first, then 0.	8410 and 1048.
Make a list in order: 3-digit numbers from 0, 3, 6. Fix the first digit: 3, then 6 (never 0). For each, list the rest from smallest.	4 numbers: 306, 360, 603, 630.

SAY *"The key idea today is: start at the biggest place. Decide the thousands digit first. To find every number, fix the first digit and list the rest in order. Say it with me."*

DO Show the key idea slide at minutes 34 to 35. The "common mistakes" on that slide are for you, not the children. Watch for them during the worksheet:

- Putting 0 first for the smallest number (0357 is really 357, a 3-digit number).
- For the greatest odd number, making the greatest number first and then trying to fix it, instead of choosing the ones digit first.
- Listing numbers in a random order, so some are missed or written twice (especially with the two 5s in Q2).

35–40 min Brain break

DO Place value moves. The tutor says a 4-digit number: jump for each thousand, clap for each hundred, tap your knees for each ten, touch your toes for each one.

Water. Toilet break for anyone who needs it. Keep it to five minutes.

DO Story break (optional, 2 minutes): after the movement, show the Story break slide (Rozsa Peter, Hungary (Budapest), 1905 to 1977) and read the story aloud. Link to today: Today you continue number patterns, and a rule that makes each number from the one before is the idea of recursion that she studied. Then offer the try-it as a question for everyone to think about: "undefined" Anyone may pass, nobody is called on by name and nothing is marked. The answer is in the notes of the slide and in the answer key below. Skip it if you are short of time: click straight on to the worksheet.

40–72 min Worksheet (Start here pages, then your choice, with its own Investigation slot at 67–72)

DO Minute 40: give out the workbook. Read the title of Lesson 1 aloud. Say "Pairs first. Talk before you write."

Minutes 40 to 52: page 1 (Start here) in pairs; where the deck has a short teaching slide, it comes first. Pairs are in breakout rooms, so visit EVERY room by minute 47 (about a minute each; camera

on, parent nearby), then check in again at about minute 52. Minutes 52 to 54: check page 1 together. Minutes 54 to 58: short teaching before page 2. Minutes 58 to 65: each child works alone on page 2. Minutes 65 to 67: check page 2 together. Minutes 67 to 72: everyone chooses Think harder, Go further or the Investigation (the Investigation has its own slide, minutes 67 to 72). A child who finishes early starts on one of them at once, and any child may choose them once page 1 and Q4 and Q5 of page 2 are done.

SAY *"Pairs first. Read each question aloud and talk about it before you write. I am not looking for speed. I am looking for good thinking."*

DO Circulate and ask:

- Ask: "Which place is worth the most?"
- Ask: "Where does the odd digit of an odd number go?"
- Ask: "How do you know your list has every number?"

IF STUCK Hint 1: "Decide the thousands digit first."

Hint 2: "For odd or even, decide the ones digit first."

Hint 3: "Make a list: fix the first digit, then go in order from smallest."

DO Anyone who wants more may choose Think harder first, then Go further, or the Investigation (its easy first step is on page 2 of Start here). Invite the whole group: "When you have finished Start here you can choose Think harder, Go further or the Investigation." Never announce who is on which page, and do not split a pair unless both children agree.

Think harder and Go further questions

Question	Answer
Workbook Think harder 1: Use the cards 5, 0, 8, 2 once each. Make the greatest even number and the smallest odd number.	Greatest even: 8520. Smallest odd: 2085 (5 is the only odd card, so it goes last).
Workbook Think harder 2: How many 4-digit numbers can you make with the cards 1, 1, 0, 0? List them.	3 numbers: 1001, 1010, 1100.
Workbook Think harder 3: A number has 7 thousands, 13 hundreds and 6 ones. What is the number?	8306 (7000 + 1300 + 6)
Workbook Think harder 4: I am between 5000 and 6000. My hundreds, tens and ones digits are all the same. My digits add up to 11. What number am I?	5222 (5 + 2 + 2 + 2 = 11)

Workbook Go further 5: Make up your own "What number am I?" puzzle. Choose a secret 4-digit number and write 3 or 4 clues about its digits. Check that ONLY ONE number fits all your clues. Then swap with a partner.	Answers vary. Check that every clue is true for the secret number and that the clues together leave only one number (list the numbers that fit clue 1, then cross out with clue 2, and so on). Example: "I am between 4000 and 5000. My hundreds digit is 1 less than my thousands digit. My ones digit is 3 times my tens digit. My digits add up to 15." The only number is 4326.
Workbook Go further 6: A 4-digit number has all different digits. Its thousands digit is the sum of its other three digits. What is the greatest such number? What is the smallest?	Greatest 9810 (9 = 8 + 1 + 0). Smallest 3012 (the smallest sum of three different digits is 0 + 1 + 2 = 3).
Workbook Go further 7: Mei uses the cards 1, 2, 3, 4 once each to make 4-digit numbers. She lists them all from smallest to greatest. What is the 10th number in her list?	2341. Six numbers start with 1; then 2134 (7th), 2143 (8th), 2314 (9th), 2341 (10th).

These are the questions on the Think harder and Go further pages in the workbook. More are in the separate Question Bank.

72–80 min Talk time (Investigation debrief first)

DO Bring the group back together (camera on). First 3 minutes (72 to 75): the Investigation debrief. Ask for volunteers to share what they noticed, then show the model answer (in the Investigation row at the end of this lesson's answer key and in the notes of the Investigation slide). An unfinished Investigation can be shared at the start of next lesson. Then the talk question. Ask for volunteers first. A pass is always fine, and so is sharing with a partner. Do not call a child by name cold, and keep no public tallies or rankings. Nobody is called on by name. Come back to sneak-peek puzzle 2: "Now we know the idea, how could it help?"

SAY *"Is 0357 a 4-digit number?"*

"How do you know you found ALL the numbers in your list?"

Discussion question	Listen for
Mei says the smallest 4-digit number she can make with 3, 0, 7, 5 is 0357. Do you agree?	No. 0357 is really 357, a 3-digit number, so the smallest 4-digit number is 3057. We know a list is complete because we fixed the first digit and listed the rest in order from smallest, so nothing is missed or repeated.

80–90 min Exit ticket (80–86), look back (86–88) and closing (88–90)

DO Minutes 80 to 86. Give each child a slip of paper (or ask them to write on their journal page). No talking, no helping. Show the exit ticket slide. Children show the answer to you on camera in the open call (or keep it to share if they want); no photos and no private messages. Never mark or read answers out in front of the group.

Exit ticket	Answer

Use the cards 6, 0, 3, 8 once each. Write the smallest 4-digit number and the greatest odd 4-digit number.	Smallest: 3068. Greatest odd: 8603 (3 is the only odd card, so it goes in the ones place).

Look back (minutes 86 to 88, two minutes)

SAY *"Turn to the last page of today's lesson and find Look back. Tick or write one thing for each line: something you did well, the hardest part, and what helped you. There are no right answers, and you may pass on any line."*

"Would anyone like to share something they did well, or what helped? Thank you. When you got stuck and tried something else, that is exactly what mathematicians do."

DO Give a full 2 minutes. Never collect the page, mark it or read it out in front of the group: children keep it. You cannot see the paper. Tell the WHOLE group: "If you ticked still stuck, say check-in please in the open class, or ask your grown-up to tell me." Then talk it through with that child in the open session, with a parent nearby. No hand signals that single a child out, never a private message. Open the next lesson with the whole group: "What is still tricky from last time?" Invite volunteers only. Praise the strategy ("you drew it and that got you unstuck"), not the speed or the answer.

SAY *"Sleuths use every clue and check every case. Today you made lists in order and proved you had them all."*

DO Minutes 88 to 90. Stamp the inquiry journal when a child has shown you their exit ticket. Mark S, D or H in the progress tracker at the back of this guide (it is for you only). If a child is marked D or H, plan two minutes of quiet catch-up at the start of the next lesson.

Extension for children who want more

With the cards 1, 2, 3, 4, how many 4-digit numbers can you make? How many of them are odd? (24 numbers; 12 are odd, because 6 end in 1 and 6 end in 3.)

Answer key (workbook)

Q1 (a) 7530 (b) 3057 (c) 7503 (d) 5037 (e) It is only 37 away from 5000. The closest number below 5000 is 3750, which is 1250 away.

Q2 Starting with 2: 2055, 2505, 2550. Starting with 5: 5025, 5052, 5205, 5250, 5502, 5520. (a) 9 numbers (b) 4 (5205, 5250, 5502, 5520)

Q3 (a) 9550, 9450 (take away 100) (b) 4567, 5678 (add 1111) (c) 1015, 1021 (add 1, 2, 3, 4, then 5, then 6) (d) Yes, 5000 is in the pattern (3000 + 8 jumps of 250). The number right after it is 5250

Q4 (a) 5527 (b) 13 hundreds (c) No. 23 tens is 230, so the number is 6000 + 230 = 6230.

Q5 (a) 6903 (thousands 6, ones 3, hundreds 6 + 0 + 3 = 9, so the tens digit must be 0) (b) Greatest 3201, smallest 1023. The only four different digits that add up to 6 are 0, 1, 2, 3, and an odd number ends in 1 or 3.

Q6 10 numbers. Starting with 3: 3000. Starting with 2: 2100, 2010, 2001. Starting with 1: 1200, 1110, 1101, 1020, 1011, 1002. (b) 1, 3 and 6 numbers: the counts go up by 2, then by 3.

Sneak-peek puzzles Puzzle 1: 940 Puzzle 2: 3049

Investigation (What if one card changes?) Model answer and what to look for: Starting with 1: 1023, 1032, 1203, 1230, 1302, 1320 (6 numbers). 6 start with 2 and 6 start with 3. None start with 0 (0123 is only a 3-digit number). Altogether 18, not 24. Same: each of the cards 1, 2, 3 still makes 6 numbers. Changed: 0 cannot go first, so 6 numbers are lost (24 – 6 = 18). Look for: a list in order, a count by first digit, and the reason that zero cannot lead. Extension: how many are even? (10 with the cards 0, 1, 2, 3; 12 with the cards 1, 2, 3, 4.)

Story break try-it (Rozsa Peter, optional) 11, 23, 47, 95 (5 x 2 + 1 = 11, 11 x 2 + 1 = 23, 23 x 2 + 1 = 47, 47 x 2 + 1 = 95). The next one is 95 x 2 + 1 = 191.

Go further (explore slide) Cards 2, 4, 5, 8 now: how many numbers between 4000 and 6000? Which is closest to 5000? Answer: 12 numbers: 6 start with 4 and 6 start with 5 (2, 4, 5, 8 each once). Closest to 5000: 4852 (148 away); the nearest above is 5248.

Lesson 2: Cryptarithm Crackers

Week 1 (second lesson) • Addition and Subtraction • Workbook: Lesson 2

Objective	Find missing digits in 4-digit addition and subtraction by working column by column from the ones, check subtraction by adding, work backwards in magic squares and number pyramids, and use guess, check and improve for digit-card and letter puzzles.
Thinking skills	Column addition and subtraction with regrouping, using the opposite operation (work backwards), guess and check then improve, reasoning about carrying, explaining "this is the only answer"
Materials	Digit cards 0–9 (one set per child), mini whiteboards, a printed 3 × 3 grid, counters or small paper squares to move around. For the sneak-peek puzzles: paper, a pencil and a few counters or small objects for each child.
Online set-up	Children use the cut-out digit cards at the back of the workbook (or paper squares numbered 0–9) and draw the grids on paper. Share the screen for the column sums and write on them.
Prepare	Solve the worksheet yourself first. Set out one materials box per pair. Open the slides for this lesson and check the answers in the speaker notes.

0–2 min Opening

SAY *"Welcome, Cryptarithm Crackers! Some digits in today's sums have gone missing, and some have been turned into letters. Like code-breakers, we will crack them one column at a time and prove our answer is the only one."*

DO Greet each child by name. Show the lesson title slide. Read "Today we will…" aloud.

Ask the whole group once: "What is still tricky from last time? Did anyone try the Investigation at home? Volunteers can share one thing." Pass is fine.

2–10 min Sneak-peek puzzles (2–5), warm-up game (5–8) and quick questions (8–10)

SAY *"Here are two sneak-peek puzzles. These are ideas we have not learnt yet, so guessing is welcome and so is "I do not know". Nobody is marked. Write your answer on your own paper. When I say "show me", hold it up to the camera for me."*

DO Show the "Sneak-peek puzzles" slide (minutes 2 to 5). Read puzzle 1 aloud and give about 1 minute: it is an access puzzle, so any child can try it with objects, a drawing or counting. Then puzzle 2, about 1 minute, the thinking puzzle. Look at the papers on camera quietly. Do not show the answers on the slide and do not read answers out. Say "thank you" to everyone in the same way.

Nobody is labelled. Never say who got it, who is on which page, or "fast group" or "top group". Do not single anyone out. Invite the whole group once Start here is done. It is an invitation, never a ranking: every child can choose Think harder or the Investigation once page 1 and Q4 and Q5 of page 2 are done. We come back to puzzle 2 in talk time, for volunteers.

Sneak-peek puzzle	Answer (tutor only) and what to listen for
Puzzle 1 (access): Find the missing digit A: 2A5 + 134 = 389. Try each digit for A, or add the ones, then the tens, then the hundreds.	A = 5. Listen for: Tens: A + 3 must make 8, so A = 5. Check: 255 + 134 = 389.
Puzzle 2 (thinking): Ali had some marbles. He won 1250 more and then had 4000. How many did he have at first? Draw a number line if it helps.	2750. Listen for: Work backwards: take away what he won. 4000 − 1250 = 2750. Check: 2750 + 1250 = 4000.

Choice during the lesson (groups of 8, online)

- Minutes 10 to 40 (explore, mini-lesson, brain break): everyone does the same. The explore slide has a Go further what-if from minute 10, so a pair that has finished its first task has real work at once. Do not say who is ready for it.
- Minute 40: everyone opens Start here page 1 and works in pairs. A line at the top says: “Do page 1, then Q4 and Q5 on page 2. After that you may choose: Think harder, Go further or the Investigation.” You never announce who is on which page. Invite the whole group, never one child: “When you have finished Start here you can choose Think harder, Go further or the Investigation.” Do not split a pair unless both children agree.
- The first-step help for the Investigation is printed on page 2 of Start here, so every child sees it. Pairs work in breakout rooms, so visit EVERY room by minute 47 (about a minute each, camera on, a parent nearby) and ask “how do you know?”, not “is it done?”. At about minute 52 check in again with two or three children, including children on Start here.
- Minute 67: everyone chooses what to do next: finish page 2, Think harder, Go further or the Investigation. The Investigation has its own slot, minutes 67 to 72, and its debrief opens talk time. Everyone is back together for talk time at minute 72. An Investigation that is not finished goes home; volunteers share one thing at the start of next lesson.
- Next lesson everything starts again from the same pages: nobody is on a track.

DO Warm-up game (minutes 5 to 8). Now the game, then the quick questions (minutes 8 to 10): Make 1000. The tutor shows a number; children write the number that makes 1000 with it: 350, 720, 105, 999, 460, 75.

SAY *“I show you a number. You write the number that makes 1000 with it. Ready? 350.”*

“How did you work it out? Did anyone go to 400 first, then to 1000?”

DO After the game, ask the quick questions on the warm-up slides. Use thumbs or a whisper to partners before taking answers.

Ask	Expect
3456 + 1000 = ?	4456
5000 − 1 = ?	4999
2700 + 300 = ?	3000

8000 – 2500 = ?	5500
4999 + 2 = ?	5001

10–25 min Explore (in pairs, with objects)

SAY *"Can you use the digits 1 to 9, each once, to make a sum of two 3-digit numbers with a 3-digit answer? Do you think there are many ways, a few, or none?"*

DO 1. Take the digit cards 1 to 9.

2. Make two 3-digit numbers that add up to a 3-digit number, using every card once.

3. Check the sum by adding.

4. Find more. What is always true?

SAY *"Talk to your partner while you work. I will come round and ask you questions."*

IF STUCK Stuck: "Put small digits in the hundreds of the two numbers you add. Try 1 and 2 in the hundreds." (This is also printed on the explore slide.)

Close but a digit repeats: "Swap two cards in the ones column and add again."

Ready for more: the Go further what-if on the slide (smallest and biggest answer). Or: "Add up the digits of your answer. Does a friend's answer give the same total?"

DO Pairs work from minute 10. If a pair finishes the first task, they may take the Go further what-if on the slide (do not say who). At about minute 15 bring the group back together and take the three questions below, one at a time (minutes 15 to 25).

Ask	Expect
How many sums did you find?	There are many: 168 if 173 + 286 and 286 + 173 count once. For example 173 + 286 = 459 and 218 + 349 = 567.
How many times did you carry?	Exactly once in every sum.
Add the digits of your answer. What do you notice?	They always add up to 18 (4 + 5 + 9, 5 + 6 + 7).

25–35 min Mini-lesson (you teach, they join in)

DO Show the mini-lesson slides. Think aloud as you work. Say what you notice, what you test and how you check. Children copy how you think.

SAY *"4A68 + 17B5 = C043. I start at the ones: 8 + 5 = 13. I write 3 and carry 1."*

"Tens: 6 + B + the carried 1 must end in 4. 6 + 7 + 1 = 14. So B is 7, and I carry 1 again."

"Hundreds: A + 7 + 1 must end in 0, so it makes 10. A is 2. Thousands: 4 + 1 + 1 = 6, so C is 6. So 4268 + 1775 = 6043. I check by adding it all again."

"For subtraction I turn it round: 6A02 – B35C = 1947 means 1947 + B35C = 6A02. Now I can add, column by column."

Worked example on the slides	Answer
Missing digits: 4A68 + 17B5 = C043. Ones: 8 + 5 = 13, write 3, carry 1. Tens: 6 + B + 1 = 14, so B = 7, carry 1. Hundreds: A + 7 + 1 = 10, so A = 2, carry 1. Thousands: 4 + 1 + 1 = 6, so C = 6.	4268 + 1775 = 6043.
Check by adding: 6A02 − B35C = 1947. Turn it round: 1947 + B35C = 6A02. Ones: 7 + C ends in 2, so C = 5 (12), carry 1. Tens: 4 + 5 + 1 = 10, carry 1. Hundreds: 9 + 3 + 1 = 13, so A = 3, carry 1. Thousands: 1 + B + 1 = 6, so B = 4.	6302 − 4355 = 1947.

SAY *"The key idea today is: one column at a time. Start at the ones and carry to the left. Check subtraction by adding. Stuck? Guess, check, then improve. Say it with me."*

DO Show the key idea slide at minutes 34 to 35. The "common mistakes" on that slide are for you, not the children. Watch for them during the worksheet:

- Forgetting the carried 1 when working out the missing digit in the next column.
- In subtraction, looking for a digit that can be taken from 3 to leave 8 (there is none) instead of regrouping: 13 − 5 = 8.
- Magic squares: guessing numbers instead of finding a full line first to get the total.

35–40 min Brain break

DO Code-breaker moves. Tutor says a digit: odd means hop on one foot that many times, even means clap that many times. Then 0: freeze!

Water. Toilet break for anyone who needs it. Keep it to five minutes.

DO Story break (optional, 2 minutes): after the movement, show the Story break slide (Joan Clarke, England (London, Cambridge, Bletchley Park), 1917 to 1996) and read the story aloud. Link to today: Today you crack letter puzzles by trying possibilities and working out what must be true, and codebreakers also cut down possibilities to find the secret. Then offer the try-it as a question for everyone to think about: "undefined" Anyone may pass, nobody is called on by name and nothing is marked. The answer is in the notes of the slide and in the answer key below. Skip it if you are short of time: click straight on to the worksheet.

DO Accuracy note for you (do not read out): Sources differ slightly on the year she became deputy head and on her honours, so the story gives no years for those. Letter puzzles are our link; the sources do not say she did school-style puzzles.

40–72 min Worksheet (Start here pages, then your choice, with its own Investigation slot at 67–72)

DO Minute 40: give out the workbook. Read the title of Lesson 2 aloud. Say "Pairs first. Talk before you write."

Minutes 40 to 52: page 1 (Start here) in pairs; where the deck has a short teaching slide, it comes first. Pairs are in breakout rooms, so visit EVERY room by minute 47 (about a minute each; camera on, parent nearby), then check in again at about minute 52. Minutes 52 to 54: check page 1

together. Minutes 54 to 58: short teaching before page 2. Minutes 58 to 65: each child works alone on page 2. Minutes 65 to 67: check page 2 together. Minutes 67 to 72: everyone chooses Think harder, Go further or the Investigation (the Investigation has its own slide, minutes 67 to 72). A child who finishes early starts on one of them at once, and any child may choose them once page 1 and Q4 and Q5 of page 2 are done.

SAY *"Pairs first. Read each question aloud and talk about it before you write. I am not looking for speed. I am looking for good thinking."*

DO Circulate and ask:

- Ask: "Which column did you start with? Why?"
- Ask: "Is there a carry coming into this column?"
- Ask: "How do you know this is the ONLY answer?"

IF STUCK Hint 1: "Start at the ones column."

Hint 2: "Write the carried 1 above the next column."

Hint 3: "For subtraction, check by adding."

DO Anyone who wants more may choose Think harder first, then Go further, or the Investigation (its easy first step is on page 2 of Start here). Invite the whole group: "When you have finished Start here you can choose Think harder, Go further or the Investigation." Never announce who is on which page, and do not split a pair unless both children agree.

Think harder and Go further questions

Question	Answer
Workbook Think harder 1: Find the missing digits (turn it round and add): 7A03 − B46C = 2538	A = 0, B = 4, C = 5: 7003 − 4465 = 2538 (check: 2538 + 4465 = 7003)
Workbook Think harder 2: Find the missing digits: 2A7B + C6D9 = 6000	A = 3, B = 1, C = 3, D = 2: 2371 + 3629 = 6000 (the only way)
Workbook Think harder 3: In a number pyramid, each brick is the sum of the two bricks under it. The bottom row is 1200, A, B. The middle row is 2650, C. The top brick is 6000. Find the missing bricks A, B and C.	Bottom row 1200, 1450, 1900. Middle row 2650, 3350.
Workbook Think harder 4: Ali meant to add 1099 to a number, but he added 1999 by mistake and got 6543. What is the correct answer?	5643 (he added 900 too much: 6543 − 900)

Workbook Go further 5: Make up your own missing-digits sum. Write a 4-digit + 4-digit addition that works, then rub out exactly 3 digits in 3 DIFFERENT columns and write the letters A, B and C instead. Swap with a partner. Does your partner find only one answer? Explain why your sum has only one answer.	Answers vary. Check that the sum is true and that the letters are in different columns, so that each column fixes one digit (3 letters in the same column, or a letter in the answer that can be anything, gives many answers). Example: 3517 + 2486 = 6003, rubbed out to A517 + 2B86 = 60C3. The only answer is 3517 + 2486 = 6003.
Workbook Go further 6: Use the digit cards 1 to 8 once each to make two 4-digit numbers. Make their difference as small as possible.	247: 5123 – 4876 (the only way). The thousands digits must be next to each other; 5 and 4 in the middle work best.
Workbook Go further 7: ABC + ACB = CBA. Different letters are different digits. Find them.	A = 4, B = 5, C = 9: 459 + 495 = 954

These are the questions on the Think harder and Go further pages in the workbook. More are in the separate Question Bank.

72–80 min Talk time (Investigation debrief first)

DO Bring the group back together (camera on). First 3 minutes (72 to 75): the Investigation debrief. Ask for volunteers to share what they noticed, then show the model answer (in the Investigation row at the end of this lesson's answer key and in the notes of the Investigation slide). An unfinished Investigation can be shared at the start of next lesson. Then the talk question. Ask for volunteers first. A pass is always fine, and so is sharing with a partner. Do not call a child by name cold, and keep no public tallies or rankings. Nobody is called on by name. Come back to sneak-peek puzzle 2: "Now we know the idea, how could it help?"

SAY *"Why do we start at the ones column?"*

"How can a puzzle have only one answer?"

Discussion question	Listen for
Why do we always start a missing-digit puzzle at the ones column?	A carry moves from right to left, so the ones column is the only column with nothing coming into it. Once we know the ones, we know the carry, and then the next column has only one possible digit, and so on.

80–90 min Exit ticket (80–86), look back (86–88) and closing (88–90)

DO Minutes 80 to 86. Give each child a slip of paper (or ask them to write on their journal page). No talking, no helping. Show the exit ticket slide. Children show the answer to you on camera in the open call (or keep it to share if they want); no photos and no private messages. Never mark or read answers out in front of the group.

Exit ticket	Answer
Find the missing digit A: 5A3 + 126 = 689	563 + 126 = 689. A = 6 (ones 3 + 6 = 9; tens A + 2 = 8, so A = 6).

Look back (minutes 86 to 88, two minutes)

SAY *"Turn to the last page of today's lesson and find Look back. Tick or write one thing for each line: something you did well, the hardest part, and what helped you. There are no right answers, and you may pass on any line."*

"Would anyone like to share something they did well, or what helped? Thank you. When you got stuck and tried something else, that is exactly what mathematicians do."

DO Give a full 2 minutes. Never collect the page, mark it or read it out in front of the group: children keep it. You cannot see the paper. Tell the WHOLE group: "If you ticked still stuck, say check-in please in the open class, or ask your grown-up to tell me." Then talk it through with that child in the open session, with a parent nearby. No hand signals that single a child out, never a private message. Open the next lesson with the whole group: "What is still tricky from last time?" Invite volunteers only. Praise the strategy ("you drew it and that got you unstuck"), not the speed or the answer.

SAY *"Code-breakers check every column. Today you worked backwards, guessed and improved, and proved your answers."*

DO Minutes 88 to 90. Stamp the inquiry journal when a child has shown you their exit ticket. Mark S, D or H in the progress tracker at the back of this guide (it is for you only). If a child is marked D or H, plan two minutes of quiet catch-up at the start of the next lesson.

Extension for children who want more

Find three different sums that add two 3-digit numbers to make a 3-digit answer, using the digits 1 to 9 once each. Add the digits of each answer. What do you notice? (They always add up to 18, e.g. 173 + 286 = 459 and 218 + 349 = 567.)

Answer key (workbook)

Q1 (a) 532 − 216 = 316 (the missing digit is 3) (b) 3447 + 2578 = 6025 (c) 316 + 216 = 532

Q2 (a) 1173, e.g. 642 + 531 (b) 381, e.g. 135 + 246 (c) 47: 412 − 365 (the only way)

Q3 (a) Magic total 4500. Middle row 1900, 1500, 1100. Bottom row 1400, 1300, 1800. (b) Numbers opposite each other across the centre add up to 3000 (e.g. 1200 + 1800, 1700 + 1300).

Q4 (a) Middle row 2375, 3600. Bottom row 1250, 1125, 2475. (b) Both corner bricks are 1500 (middle row 3500, 3500).

Q5 (a) 5872 and 7406 (b) The two digits in each place add up to 9, so there is never a carry. (c) 5873 (one more than 9999 − 4127 = 5872)

Q6 A = 5, B = 0, C = 1: 50 + 50 + 50 = 150

Sneak-peek puzzles Puzzle 1: A = 5 Puzzle 2: 2750

Investigation (Is it always true? The reverse trick) Model answer and what to look for: Every answer is in the 99 times table: 99 × (first digit − last digit). 842 − 248 = 594 = 99 × 6 (8 − 2 = 6); 720 − 27 = 693 = 99 × 7. This is true for every 3-digit number whose first digit is bigger than its last (checked by computer for all of them). Why: the middle digit is in the same place in both numbers, so it cancels. The hundreds and ones swap places, so each step of difference between the first and last digits adds 99

(100 − 1). Look for: several tests, spotting the 99s (not only 495; 495 appears only when the digits differ by 5), and any reason about the middle digit. Extension: when is the answer exactly 495?

Story break try-it (Joan Clarke, optional) A = 5 and B = 1, because 5 + 5 + 5 = 15 and 15 reads B A (1 then 5). No other pair of different digits works.

Go further (explore slide) With the cards 1 to 9, what is the smallest answer you can make? The biggest? Answer: Smallest 459 (for example 173 + 286); biggest 981 (for example 235 + 746). Every answer has digits adding to 18.

Lesson 3: Times Table Treasure

Week 2 (first lesson) • Tables of 6, 7, 8 and 9 • Workbook: Lesson 3

Objective	Find and explain patterns in the 6, 7, 8 and 9 times tables, use last-digit patterns, split an array into facts we know, and solve grid and list puzzles in a careful order.
Thinking skills	Look for a pattern, make a systematic list, use known facts, explain why
Materials	Multiplication square (1 to 10), dot paper, coloured pencils, a ruler, mini whiteboards. For the sneak-peek puzzles: paper, a pencil and a few counters or small objects for each child.
Online set-up	Tutor draws on a shared digit-circle slide. Each child draws 10 dots round a circle (like a clock) labelled 0 to 9, and has dot paper and coloured pencils.
Prepare	Solve the worksheet yourself first. Set out one materials box per pair. Open the slides for this lesson and check the answers in the speaker notes.

0–2 min Opening

SAY *"Today we are treasure hunters inside the times tables. The 6, 7, 8 and 9 tables are full of hidden patterns. If you can spot a pattern, you can use it to check answers and to find answers you have never learnt."*

DO Greet each child by name. Show the lesson title slide. Read "Today we will…" aloud.

Ask the whole group once: "What is still tricky from last time? Did anyone try the Investigation at home? Volunteers can share one thing." Pass is fine.

2–10 min Sneak-peek puzzles (2–5), warm-up game (5–8) and quick questions (8–10)

SAY *"Here are two sneak-peek puzzles. These are ideas we have not learnt yet, so guessing is welcome and so is "I do not know". Nobody is marked. Write your answer on your own paper. When I say "show me", hold it up to the camera for me."*

DO Show the "Sneak-peek puzzles" slide (minutes 2 to 5). Read puzzle 1 aloud and give about 1 minute: it is an access puzzle, so any child can try it with objects, a drawing or counting. Then puzzle 2, about 1 minute, the thinking puzzle. Look at the papers on camera quietly. Do not show the answers on the slide and do not read answers out. Say "thank you" to everyone in the same way.

Nobody is labelled. Never say who got it, who is on which page, or "fast group" or "top group". Do not single anyone out. Invite the whole group once Start here is done. It is an invitation, never a ranking: every child can choose Think harder or the Investigation once page 1 and Q4 and Q5 of page 2 are done. We come back to puzzle 2 in talk time, for volunteers.

Sneak-peek puzzle	Answer (tutor only) and what to listen for

Puzzle 1 (access): Here is the 7 times table: 7, 14, 21, 28, 35, 42, 49, 56, 63, 70. Which of these numbers are in it: 49, 54, 63?	49 and 63. Listen for: Look along the table: 49 and 63 are there; 54 is not.
Puzzle 2 (thinking): You know 6 × 7 = 42. Use it to work out 6 × 14 and 6 × 6. Say how you did it.	84 and 36. Listen for: 14 is double 7, so double 42 = 84. 6 × 6 is one 6 less than 6 × 7: 42 − 6 = 36.

Choice during the lesson (groups of 8, online)

- Minutes 10 to 40 (explore, mini-lesson, brain break): everyone does the same. The explore slide has a Go further what-if from minute 10, so a pair that has finished its first task has real work at once. Do not say who is ready for it.
- Minute 40: everyone opens Start here page 1 and works in pairs. A line at the top says: “Do page 1, then Q4 and Q5 on page 2. After that you may choose: Think harder, Go further or the Investigation.” You never announce who is on which page. Invite the whole group, never one child: “When you have finished Start here you can choose Think harder, Go further or the Investigation.” Do not split a pair unless both children agree.
- The first-step help for the Investigation is printed on page 2 of Start here, so every child sees it. Pairs work in breakout rooms, so visit EVERY room by minute 47 (about a minute each, camera on, a parent nearby) and ask “how do you know?”, not “is it done?”. At about minute 52 check in again with two or three children, including children on Start here.
- Minute 67: everyone chooses what to do next: finish page 2, Think harder, Go further or the Investigation. The Investigation has its own slot, minutes 67 to 72, and its debrief opens talk time. Everyone is back together for talk time at minute 72. An Investigation that is not finished goes home; volunteers share one thing at the start of next lesson.
- Next lesson everything starts again from the same pages: nobody is on a track.

DO Warm-up game (minutes 5 to 8). Now the game, then the quick questions (minutes 8 to 10): Buzz! Count round the group from 1 to 40 with the 7s strip on the screen. Hold up an open palm (Buzz) instead of saying a number in the 7 times table. Only the 7s are played.

SAY *“We are going to count round the group up to 40. The 7s strip is on the screen. When it is a number in the 7 times table, do not say it: hold up an open palm. That is a Buzz! If anyone misses one, that is fine: it is a tricky one and we check on the strip.”*

“Which numbers did we Buzz? What do you notice about them?”

DO After the game, ask the quick questions on the warm-up slides. Use thumbs or a whisper to partners before taking answers.

Ask	Expect
6 × 7 = ?	42
9 × 9 = ?	81
8 × ? = 64	8

Which is bigger: 6 × 8 or 7 × 7?	7 × 7 = 49 is 1 more than 6 × 8 = 48

10–25 min Explore (in pairs, with objects)

SAY *"Here is a circle with the digits 0 to 9, like a clock. I write the 6 times table and look only at the last digit: 6, 12, 18 ... the last digits are 6, 2, 8. If I join those dots in order, what shape do you think we will get? Will it ever touch an odd digit?"*

DO 1. Draw 10 dots round a circle. Label them 0 to 9, like a clock.

2. Write the 6 times table. Circle the last digit of each answer.

3. Join the last digits in order: 6 → 2 → 8 → 4 → 0 → back to 6.

4. Now do the same for the 4s and the 8s. Compare the shapes.

SAY *"Talk to your partner while you work. I will come round and ask you questions."*

IF STUCK Stuck: write the table down the side of your page first, then circle the last digit of each answer.

Confused: "Look only at the ones digit. 6 × 3 = 18, so we join to the 8."

Ready for more: the Go further what-if on the slide (the 5s and the 9s). Or try the 7s and the 3s and compare.

DO Pairs work from minute 10. If a pair finishes the first task, they may take the Go further what-if on the slide (do not say who). At about minute 15 bring the group back together and take the three questions below, one at a time (minutes 15 to 25).

Ask	Expect
Which tables made the same shape?	The 4s and the 6s make the same star (going round opposite ways). The 2s and the 8s make the same shape.
Why does the 6s shape never touch an odd digit?	Every answer in the 6 times table is even: 6 groups can be put into 3 pairs of groups.
Which table do you think makes the same shape as the 3s?	The 7s, because 3 + 7 = 10, just like 4 + 6 and 2 + 8.

25–35 min Mini-lesson (you teach, they join in)

DO Show the mini-lesson slides. Think aloud as you work. Say what you notice, what you test and how you check. Children copy how you think.

SAY *"Look at the 8 times table: 8, 16, 24, 32, 40, 48 ... The last digits go 8, 6, 4, 2, 0, then 8, 6, 4, 2, 0 again. So the pattern repeats every 5."*

"What is the last digit of 8 × 23? 8 × 20 ends in 0, so only 8 × 3 matters. 8 × 3 = 24, so 8 × 23 ends in 4. Check: 8 × 23 = 184. Yes."

"I do not know 6 × 8 by heart? I split the array: 5 rows of 8 is 40, one more row of 8 is 8, so 6 × 8 = 48. For 9 × 7 I can do 10 sevens, which is 70, and take one 7 away: 63."

Worked example on the slides	Answer
Last digits repeat. The 8 times table ends in 8, 6, 4, 2, 0 and then repeats. What is the last digit of 8 × 23? 8 × 20 ends in 0, so only 8 × 3 = 24 matters.	4 (8 × 23 = 184).
Split the array. 9 × 7: split 9 rows of 7 into 5 rows and 4 rows. 5 × 7 = 35 and 4 × 7 = 28. Or: 10 × 7 = 70, take away one 7.	63.

SAY *"The key idea today is: look for a pattern. Look for a pattern and test it. Use facts you know to find facts you do not know. Say it with me."*

DO Show the key idea slide at minutes 33 to 35. The "common mistakes" on that slide are for you, not the children. Watch for them during the worksheet:

- Checking the 9s table only by "the digits add up to 9" (81 passes that test, but 9 × 8 is 72).
- Splitting an array and then giving one part as the answer (writing 40 for 7 × 8 instead of 40 + 16).
- Missing or repeating ways in a list because the rows are not tried in order 2, 3, 4, 5 …

35–40 min Brain break

DO Times-table jumps. Tutor calls a fact like 6 × 3: children do that many star jumps in groups of 6, counting 6, 12, 18 aloud.

Water. Toilet break for anyone who needs it. Keep it to five minutes.

DO Story break (optional, 2 minutes): after the movement, show the Story break slide (John Napier, Scotland, 1550 to 1617) and read the story aloud. Link to today: Napier's rods are made of times tables, so they turn multiplication into reading a table and adding, just like the tables of 6, 7, 8 and 9 we explore today. Then offer the try-it as a question for everyone to think about: "undefined" Anyone may pass, nobody is called on by name and nothing is marked. The answer is in the notes of the slide and in the answer key below. Skip it if you are short of time: click straight on to the worksheet.

40–72 min Worksheet (Start here pages, then your choice, with its own Investigation slot at 67–72)

DO Minute 40: give out the workbook. Read the title of Lesson 3 aloud. Say "Pairs first. Talk before you write."

Minutes 40 to 52: page 1 (Start here) in pairs; where the deck has a short teaching slide, it comes first. Pairs are in breakout rooms, so visit EVERY room by minute 47 (about a minute each; camera on, parent nearby), then check in again at about minute 52. Minutes 52 to 54: check page 1 together. Minutes 54 to 58: short teaching before page 2. Minutes 58 to 65: each child works alone on page 2. Minutes 65 to 67: check page 2 together. Minutes 67 to 72: everyone chooses Think harder, Go further or the Investigation (the Investigation has its own slide, minutes 67 to 72). A child who finishes early starts on one of them at once, and any child may choose them once page 1 and Q4 and Q5 of page 2 are done.

SAY *"Pairs first. Read each question aloud and talk about it before you write. I am not looking for speed. I am looking for good thinking."*

DO Circulate and ask:

- Ask: "What pattern do you notice? Does it work every time? Test one more."
- Ask: "Which facts do you already know that could help you?"
- Ask: "How do you know you have found every way? What order did you go in?"

IF STUCK Hint 1: "Write the table in order and look at the ones digits only."

Hint 2: "Split the big fact into two smaller facts you know, then add."

Hint 3: "For a grid puzzle, start with a number that can only be made in one way, like 81 or 49."

DO Anyone who wants more may choose Think harder first, then Go further, or the Investigation (its easy first step is on page 2 of Start here). Invite the whole group: "When you have finished Start here you can choose Think harder, Go further or the Investigation." Never announce who is on which page, and do not split a pair unless both children agree.

Think harder and Go further questions

Question	Answer
Workbook Think harder 1: The digits of every answer in the 9 times table (9 × 1 to 9 × 10) add up to 9. Which of these are in the 9 times table: 54, 65, 72, 83?	54 and 72 (5 + 4 = 9 and 7 + 2 = 9; 6 + 5 and 8 + 3 make 11). 54 = 9 × 6, 72 = 9 × 8.
Workbook Think harder 2: What is the last digit of 8 × 8 × 8? Use last digits only.	2. 8 × 8 = 64 ends in 4, and 4 × 8 = 32 ends in 2. (8 × 8 × 8 = 512.)
Workbook Think harder 3: Ravi says: "Every answer in the 6 times table is even." Is he right? Explain why.	Yes. 6 groups can always be put into 3 pairs of groups, so nothing is left over. (6 × 7 = 3 × 14 = 42.)
Workbook Think harder 4: Find ALL the 2-digit numbers that can be divided exactly by 7 and whose digits add up to 10.	28 and 91. (List 14, 21, 28 … 98 and add the digits of each: only 28 and 91 make 10.)
Workbook Go further 5: Make up your own riddle: "I am a 2-digit number in the ___ times table …" Write 2 or 3 clues (about odd or even, the digits, being between two numbers). Check that only ONE number fits, by listing the table. Swap with a partner.	Answers vary. Check that the secret number is in the table, every clue is true, and listing the table leaves exactly one number. Example: "I am a 2-digit number in the 7 times table. I am odd. My digits add up to 10." The 2-digit odd numbers in the 7 table are 21, 35, 49, 63, 77 and 91, and only 91 has digits that add up to 10.
Workbook Go further 6: Ben keeps multiplying by 7: 7, 49, 343 … He writes only the last digit each time: 7, 9, 3 … What is the last digit when he has multiplied 8 sevens together? Explain the pattern.	1. The last digits go 7, 9, 3, 1 and then repeat, so the 4th and the 8th are both 1.

Workbook Go further 7: I am a 3-digit number less than 200. I can be divided exactly by 7 and by 9. My digits add up to 9. What number am I?	126. Numbers that work for 7 and 9 are 63, 126, 189 ...; the 3-digit ones below 200 are 126 and 189, and only 1 + 2 + 6 = 9.

These are the questions on the Think harder and Go further pages in the workbook. More are in the separate Question Bank.

72–80 min Talk time (Investigation debrief first)

DO Bring the group back together (camera on). First 3 minutes (72 to 75): the Investigation debrief. Ask for volunteers to share what they noticed, then show the model answer (in the Investigation row at the end of this lesson's answer key and in the notes of the Investigation slide). An unfinished Investigation can be shared at the start of next lesson. Then the talk question. Ask for volunteers first. A pass is always fine, and so is sharing with a partner. Do not call a child by name cold, and keep no public tallies or rankings. Nobody is called on by name. Come back to sneak-peek puzzle 2: "Now we know the idea, how could it help?"

SAY *"Why is every answer in the 6 times table even?"*

"Why does the 7 times table have odd and even answers?"

Discussion question	Listen for
Why is every answer in the 6 times table even, but the 7 times table has odd and even answers?	6 groups can always be put into 3 pairs of groups, so nothing is left over: the answer is even. With 7 groups one group has no partner, so when each group is odd the answer is odd.

80–90 min Exit ticket (80–86), look back (86–88) and closing (88–90)

DO Minutes 80 to 86. Give each child a slip of paper (or ask them to write on their journal page). No talking, no helping. Show the exit ticket slide. Children show the answer to you on camera in the open call (or keep it to share if they want); no photos and no private messages. Never mark or read answers out in front of the group.

Exit ticket	Answer
What is the last digit of 6 × 47? Explain how you know without working out 6 × 47.	2. Only 6 × 7 = 42 decides the last digit, because 6 × 40 ends in 0. (6 × 47 = 282.)

Look back (minutes 86 to 88, two minutes)

SAY *"Turn to the last page of today's lesson and find Look back. Tick or write one thing for each line: something you did well, the hardest part, and what helped you. There are no right answers, and you may pass on any line."*

"Would anyone like to share something they did well, or what helped? Thank you. When you got stuck and tried something else, that is exactly what mathematicians do."

DO Give a full 2 minutes. Never collect the page, mark it or read it out in front of the group: children keep it. You cannot see the paper. Tell the WHOLE group: "If you ticked still stuck, say check-in

please in the open class, or ask your grown-up to tell me." Then talk it through with that child in the open session, with a parent nearby. No hand signals that single a child out, never a private message. Open the next lesson with the whole group: "What is still tricky from last time?" Invite volunteers only. Praise the strategy ("you drew it and that got you unstuck"), not the speed or the answer.

SAY *"You found patterns, tested them and explained why they work. That is exactly what mathematicians do."*

DO Minutes 88 to 90. Stamp the inquiry journal when a child has shown you their exit ticket. Mark S, D or H in the progress tracker at the back of this guide (it is for you only). If a child is marked D or H, plan two minutes of quiet catch-up at the start of the next lesson.

Extension for children who want more

Draw last-digit circles for all the tables from 1 to 9 and group the tables that make the same shape. Explain why 1 and 9, 2 and 8, 3 and 7, 4 and 6 pair up.

Answer key (workbook)

Q1 Table: 9, 18, 27, 36, 45, 54, 63, 72, 81, 90 (a) The tens digit goes up by 1 and the ones digit goes down by 1; the two digits add up to 9; the tens digit is 1 less than the number of nines. (b) In 9 × 8 the tens digit is 1 less than 8, so it is 7, and 7 + 2 = 9, so 9 × 8 = 72. 81 is 9 × 9.

Q2 Answer row (given): 7, 14, 21, 28, 35, 42, 49, 56, 63, 70. Last digits: 7, 4, 1, 8, 5, 2, 9, 6, 3, 0: all ten digits, each once (a) 3 and 9 (the 3s: 3, 6, 9, 2, 5, 8, 1, 4, 7, 0; the 9s: 9, 8, 7 … 0). The 6s give only even last digits (and the 2s, 4s and 8s do too); the 5s give only 5 and 0. (b) 2, because 7 × 6 = 42 (7 × 46 = 322).

Q3 (a) 5 × 8 = 40, 2 × 8 = 16, so 7 × 8 = 56 (b) For example 5 × 6 + 4 × 6 = 30 + 24 = 54, or 10 × 6 − 6 = 54.

Q4 Rows 7, 9, 6. Columns 8, 6, 9. Row 7: 56, 42, 63. Row 9: 72, 54, 81. Row 6: 48, 36, 54. (Start with 81 = 9 × 9.)

Q5 (a) 2 rows of 18, 3 of 12, 4 of 9, 6 of 6, 9 of 4, 12 of 3, 18 of 2 (b) 7 ways (c) 6 rows of 6: turned round it is the same, so it has no partner. That is why the number of ways is odd.

Q6 (a) 24, 48, 72, 96 (b) 48 (digit sums: 24 → 6, 48 → 12, 72 → 9, 96 → 15)

Sneak-peek puzzles Puzzle 1: 49 and 63 Puzzle 2: 84 and 36

Investigation (Is it always true? Digit sums) Model answer and what to look for: 9 times table: the digit sum is 9 for 9 × 1 to 9 × 10 and for 9 × 12 = 108, but 9 × 11 = 99 gives 9 + 9 = 18. So 'always 9' is false, but 18 is also in the 9 times table (and 1 + 8 = 9 again). 3 and 6 times tables: every digit sum is in the 3 times table (3, 6, 9, 12, …). 4 times table: digit sums 4, 8, 3, 7, 2, 6, 10 … include numbers that are not in the 3 times table (4, 8), so it is not the same. Conclusion: the digit sum of any answer in the 3, 6 or 9 table is in the 3 times table; for the 9 table it is in the 9 times table. Look for: testing past × 10, finding the counterexample 99, and revising the claim instead of just saying 'no'. Do not expect a proof.

Story break try-it (John Napier, optional) 09, 18, 27, 36, 45, 54, 63, 72, 81. The tens digits go up by 1 each time (0 to 8) and the ones digits go down by 1 (9 to 1). Each pair of digits adds to 9.

Go further (explore slide) Try the 5s and the 9s. How many dots does each table touch? Why? Answer: The 5s touch only 2 dots (5 and 0). The 9s touch all 10 dots, going round the other way, like the 1s, because each 9 times answer is 1 less in the ones place than the one before.

Lesson 4: Model Makers

Week 2 (second lesson) • Word Problems: Bar Models • Workbook: Lesson 4

Objective	Draw bar models for part-whole, comparison, times-as-many and before-and-after word problems, and use them to solve two-step problems and check the answers.
Thinking skills	Draw a model, before and after, look for what stays the same, check the answer
Materials	Multilink cubes (about 30 per pair), strips of squared paper, scissors, mini whiteboards and markers. For the sneak-peek puzzles: paper, a pencil and a few counters or small objects for each child.
Online set-up	Tutor builds the bars on a shared slide. Each child uses Lego bricks, coins or paper strips at home, and a whiteboard or paper to draw bars. Print the workbook before the lesson. Pairs work in one breakout room for 10 minutes, then return for the check slide.
Prepare	Solve the worksheet yourself first. Set out one materials box per pair. Open the slides for this lesson and check the answers in the speaker notes.

0–2 min Opening

SAY *"Today we are Model Makers. Word problems can look scary, but if we draw the story as bars, the answer often jumps out at us. A good model is like a map of the problem."*

DO Greet each child by name. Show the lesson title slide. Read "Today we will…" aloud.

Ask the whole group once: "What is still tricky from last time? Did anyone try the Investigation at home? Volunteers can share one thing." Pass is fine.

2–10 min Sneak-peek puzzles (2–5), warm-up game (5–8) and quick questions (8–10)

SAY *"Here are two sneak-peek puzzles. These are ideas we have not learnt yet, so guessing is welcome and so is "I do not know". Nobody is marked. Write your answer on your own paper. When I say "show me", hold it up to the camera for me."*

DO Show the "Sneak-peek puzzles" slide (minutes 2 to 5). Read puzzle 1 aloud and give about 1 minute: it is an access puzzle, so any child can try it with objects, a drawing or counting. Then puzzle 2, about 1 minute, the thinking puzzle. Look at the papers on camera quietly. Do not show the answers on the slide and do not read answers out. Say "thank you" to everyone in the same way.

Nobody is labelled. Never say who got it, who is on which page, or "fast group" or "top group". Do not single anyone out. Invite the whole group once Start here is done. It is an invitation, never a ranking: every child can choose Think harder or the Investigation once page 1 and Q4 and Q5 of page 2 are done. We come back to puzzle 2 in talk time, for volunteers.

Sneak-peek puzzle	Answer (tutor only) and what to listen for

Puzzle 1 (access): Ali has 70 stickers and Ben has 30. Draw two bars, or use counters. How many stickers must Ali give Ben so that they have the same number?	20. Listen for: The difference is 40. Ali gives half of it: 20. Then both have 50.
Puzzle 2 (thinking): Ali has $12. Ben has $5 less than Ali. Mei has twice as much as Ben. How much money does Mei have? Draw bars to help.	$14. Listen for: Ben has 12 – 5 = $7. Mei has two bars the same as Ben's: 7 + 7 = $14.

Choice during the lesson (groups of 8, online)

- Minutes 10 to 40 (explore, mini-lesson, brain break): everyone does the same. The explore slide has a Go further what-if from minute 10, so a pair that has finished its first task has real work at once. Do not say who is ready for it.
- Minute 40: everyone opens Start here page 1 and works in pairs. A line at the top says: “Do page 1, then Q4 and Q5 on page 2. After that you may choose: Think harder, Go further or the Investigation.” You never announce who is on which page. Invite the whole group, never one child: “When you have finished Start here you can choose Think harder, Go further or the Investigation.” Do not split a pair unless both children agree.
- The first-step help for the Investigation is printed on page 2 of Start here, so every child sees it. Pairs work in breakout rooms, so visit EVERY room by minute 47 (about a minute each, camera on, a parent nearby) and ask “how do you know?”, not “is it done?”. At about minute 52 check in again with two or three children, including children on Start here.
- Minute 67: everyone chooses what to do next: finish page 2, Think harder, Go further or the Investigation. The Investigation has its own slot, minutes 67 to 72, and its debrief opens talk time. Everyone is back together for talk time at minute 72. An Investigation that is not finished goes home; volunteers share one thing at the start of next lesson.
- Next lesson everything starts again from the same pages: nobody is on a track.

DO Warm-up game (minutes 5 to 8). Now the game, then the quick questions (minutes 8 to 10): Arm bars. The tutor reads a sentence. Children show the two bars with their hands, then draw them on whiteboards in 10 seconds.

SAY *“I will read a sentence. Show me the two bars with your hands. Which hand is higher? Then draw it on your whiteboard.”*

“How did you show "3 times as many"? How did you show "altogether"?”

DO After the game, ask the quick questions on the warm-up slides. Use thumbs or a whisper to partners before taking answers.

Ask	Expect
348 + 52	400
1000 – 365	635
3 units = 27. What is 1 unit?	9

Ben has 40. Ali has 15 more than Ben. How many does Ali have?	55
Mei has 60. Siti has 12 fewer than Mei. How many does Siti have?	48

10–25 min Explore (in pairs, with objects)

SAY *"Twenty cubes make two towers. One tower has 4 more cubes than the other. Guess: how many cubes are in each tower? Write your guess, then build it to test."*

DO 1. Take 20 cubes. Build two towers. One tower must have 4 more cubes than the other.

2. Stand the towers side by side. How many cubes are in each?

3. Now use 20 cubes again: one tower must be 3 times as tall as the other.

4. Draw each pair of towers as two bars on your whiteboard.

SAY *"Talk to your partner while you work. I will come round and ask you questions."*

IF STUCK Stuck: build two towers the same height first. Then move cubes until one is 4 taller.

Confused: "Take the extra 4 cubes away first. Share what is left fairly. Then put the 4 back."

Ready for more: the Go further what-if on the slide (3 towers, 30 cubes). Or: 30 cubes where one tower is 4 times as tall as the other.

DO Pairs work from minute 10. If a pair finishes the first task, they may take the Go further what-if on the slide (do not say who). At about minute 15 bring the group back together and take the three questions below, one at a time (minutes 15 to 25).

Ask	Expect
How did you find the two towers?	Take away the extra 4 (16 left), share 16 into two equal towers of 8, then put the 4 back: 8 and 12.
How many cubes are in each tower when one is 3 times as tall?	5 and 15. Together the towers make 4 equal parts of 5.
Could you make "3 times as tall" with 22 cubes? Why?	No. We need 4 equal parts, and 22 cannot be shared into 4 equal parts (5 each leaves 2).

25–35 min Mini-lesson (you teach, they join in)

DO Show the mini-lesson slides. Think aloud as you work. Say what you notice, what you test and how you check. Children copy how you think.

SAY *"A hall has 1500 seats: 645 red, 480 blue and the rest green. It is one whole cut into parts, so I draw ONE bar for 1500 and cut it into 645, 480 and a question mark. Whole take away the parts: 1500 – 645 = 855, 855 – 480 = 375. Check: 645 + 480 + 375 = 1500."*

"Siti has 167 MORE than Ravi, so her bar is longer: Ravi's bar and an extra piece of 167. Ben has 95 FEWER than Siti, so I compare Ben with Siti, not with Ravi."

"Before and after: when both girls get the same amount, both bars grow by the same piece, so the difference stays the same. I look for what stays the same."

Worked example on the slides	Answer
Part-whole model. A hall has 1500 seats: 645 are red, 480 are blue and the rest are green. Draw one bar for 1500 and cut it into 645, 480 and ?. Whole take away the parts: 1500 – 645 – 480.	375 green seats.
Comparison model. Ravi has 236 cards. Siti has 167 more cards than Ravi. Ben has 95 fewer cards than Siti. Draw a bar for each child. Siti: 236 + 167. Ben: Siti's number – 95.	Siti 403, Ben 308.

SAY *"The key idea today is: draw a model. Draw a bar for each amount. Label what you know and put ? on what you want to find. Say it with me."*

DO Show the key idea slide at minutes 33 to 35. The "common mistakes" on that slide are for you, not the children. Watch for them during the worksheet:

- Comparing with the wrong person (Ben has 79 fewer than ALI, not Mei), or drawing "more than" as a shorter bar.
- Times as many: dividing the total by 3 instead of by 3 + 1 = 4 units.
- Make them equal: giving the whole difference instead of half of it.

35–40 min Brain break

DO Stretch and shrink. Tutor calls "longer bar": stretch your arms wide. "Shorter bar": bring them close. "3 times as long": take three big steps.

Water. Toilet break for anyone who needs it. Keep it to five minutes.

DO Story break (optional, 2 minutes): after the movement, show the Story break slide (Bhaskara II, India, 1114 to 1185) and read the story aloud. Link to today: Today we solve word problems by drawing a bar model, and Bhaskara's book is full of stories that ask us to find an unknown number. Then offer the try-it as a question for everyone to think about: "undefined" Anyone may pass, nobody is called on by name and nothing is marked. The answer is in the notes of the slide and in the answer key below. Skip it if you are short of time: click straight on to the worksheet.

DO Accuracy note for you (do not read out): A story often told says the Lilavati was named after Bhaskara's daughter (historians have no proof; it is a legend). The necklace problem in try-it is our own, written in the style of the book, not a copy of his.

40–72 min Worksheet (Start here pages, then your choice, with its own Investigation slot at 67–72)

DO Minute 40: give out the workbook. Read the title of Lesson 4 aloud. Say "Pairs first. Talk before you write."

Minutes 40 to 52: page 1 (Start here) in pairs; where the deck has a short teaching slide, it comes first. Pairs are in breakout rooms, so visit EVERY room by minute 47 (about a minute each; camera

on, parent nearby), then check in again at about minute 52. Minutes 52 to 54: check page 1 together. Minutes 54 to 58: short teaching before page 2. Minutes 58 to 65: each child works alone on page 2. Minutes 65 to 67: check page 2 together. Minutes 67 to 72: everyone chooses Think harder, Go further or the Investigation (the Investigation has its own slide, minutes 67 to 72). A child who finishes early starts on one of them at once, and any child may choose them once page 1 and Q4 and Q5 of page 2 are done.

SAY *"Pairs first. Read each question aloud and talk about it before you write. I am not looking for speed. I am looking for good thinking."*

DO Circulate and ask:

- Ask: "Is this a whole cut into parts, or two amounts being compared?"
- Ask: "Who has more? Whose bar is longer? Longer by how much?"
- Ask: "What stays the same in this story: the total or the difference?"

IF STUCK Hint 1: "Draw a bar for each person. Label what you know."

Hint 2: "Put a question mark on the part you need to find."

Hint 3: "Look for equal units. How many units are there altogether?"

DO Anyone who wants more may choose Think harder first, then Go further, or the Investigation (its easy first step is on page 2 of Start here). Invite the whole group: "When you have finished Start here you can choose Think harder, Go further or the Investigation." Never announce who is on which page, and do not split a pair unless both children agree.

Think harder and Go further questions

Question	Answer
Workbook Think harder 1: Ali and Ben have 96 cards altogether. Ali has 3 times as many cards as Ben. How many more cards does Ali have than Ben?	48. 4 units = 96, 1 unit = 24, and the difference is 2 units.
Workbook Think harder 2: Siti has 54 more beads than Mei. Siti gives Mei some beads, and now they have the same number. How many beads did Siti give Mei?	27 (half of the difference).
Workbook Think harder 3: Three boxes hold 600 pens altogether. Box A has 50 more pens than Box B. Box C has 50 more pens than Box A. How many pens are in Box C?	250. Take away the extras (50 + 100 = 150): 450 ÷ 3 = 150 in Box B, so A has 200 and C has 250.
Workbook Think harder 4: A bag of rice and a bag of flour weigh 9 kg altogether. The rice is 3 kg heavier than the flour. How heavy is each bag?	Flour 3 kg, rice 6 kg (9 − 3 = 6, 6 ÷ 2 = 3).

Workbook Go further 5: Make up your own before-and-after story about stickers or money, where one child gives some to another child. Draw its model, solve it and check it. Then change ONE number in your story. What changes in the answer, and what stays the same? Swap with a partner.	Answers vary. Check that the model matches the story, the total is the same before and after, and the child can explain what changed. Example: Ali has 50 stickers and Ben has 20. Ali gives Ben some so that they have the same number. (Ali gives 15; the difference is 30 and half of it is 15.) If Ali had 60, he would give 20: the difference grows by 10 and the gift by 5.
Workbook Go further 6: Ravi is 7 years old and his sister is 3. In how many years will their ages add up to 30? Explain how you know.	10 years. Now they add up to 10, and each year the total grows by 2 (one year for each child). 30 − 10 = 20, and 20 ÷ 2 = 10. Check: 17 + 13 = 30.
Workbook Go further 7: A ribbon 4 m 75 cm long is cut into 3 pieces. The second piece is twice as long as the first. The third piece is 25 cm longer than the second. How long is the third piece?	2 m 5 cm. 475 − 25 = 450 cm for 5 equal units, so 1 unit = 90 cm; third piece = 180 + 25 = 205 cm.

These are the questions on the Think harder and Go further pages in the workbook. More are in the separate Question Bank.

72–80 min Talk time (Investigation debrief first)

DO Bring the group back together (camera on). First 3 minutes (72 to 75): the Investigation debrief. Ask for volunteers to share what they noticed, then show the model answer (in the Investigation row at the end of this lesson's answer key and in the notes of the Investigation slide). An unfinished Investigation can be shared at the start of next lesson. Then the talk question. Ask for volunteers first. A pass is always fine, and so is sharing with a partner. Do not call a child by name cold, and keep no public tallies or rankings. Nobody is called on by name. Come back to sneak-peek puzzle 2: "Now we know the idea, how could it help?"

SAY *"Mei and Siti are each given $5. Why does the difference between them stay the same?"*

"How can a model help you check your answer?"

Discussion question	Listen for
When two children are each given the same amount of money, why does the difference between them stay the same?	Both bars grow by the same piece, so the gap between their ends does not change. To check, put your answers back into the model: the parts must make the whole and the difference must match the story.

80–90 min Exit ticket (80–86), look back (86–88) and closing (88–90)

DO Minutes 80 to 86. Give each child a slip of paper (or ask them to write on their journal page). No talking, no helping. Show the exit ticket slide. Children show the answer to you on camera in the open call (or keep it to share if they want); no photos and no private messages. Never mark or read answers out in front of the group.

Exit ticket	Answer

Ali has 3 times as many stickers as Ben. They have 80 stickers altogether. How many stickers does Ali have? Draw the model.	60. 4 units = 80, so 1 unit = 20, and Ali has 3 units.

Look back (minutes 86 to 88, two minutes)

SAY *"Turn to the last page of today's lesson and find Look back. Tick or write one thing for each line: something you did well, the hardest part, and what helped you. There are no right answers, and you may pass on any line."*

"Would anyone like to share something they did well, or what helped? Thank you. When you got stuck and tried something else, that is exactly what mathematicians do."

DO Give a full 2 minutes. Never collect the page, mark it or read it out in front of the group: children keep it. You cannot see the paper. Tell the WHOLE group: "If you ticked still stuck, say check-in please in the open class, or ask your grown-up to tell me." Then talk it through with that child in the open session, with a parent nearby. No hand signals that single a child out, never a private message. Open the next lesson with the whole group: "What is still tricky from last time?" Invite volunteers only. Praise the strategy ("you drew it and that got you unstuck"), not the speed or the answer.

SAY *"You turned words into pictures today. Next time a word problem looks hard, draw it first."*

DO Minutes 88 to 90. Stamp the inquiry journal when a child has shown you their exit ticket. Mark S, D or H in the progress tracker at the back of this guide (it is for you only). If a child is marked D or H, plan two minutes of quiet catch-up at the start of the next lesson.

Extension for children who want more

Write your own before-and-after story, draw its model and check it. Swap with a partner.

Answer key (workbook)

Q1 (a) 1250 − 486 − 378 = 386 buns (b) 486 − 378 = 108 more

Q2 (a) Ali 345 + 128 = 473; Ben 473 − 79 = 394 (b) 345 + 473 + 394 = 1212 stickers

Q3 (a) 4 units; 1 unit = 248 ÷ 4 = 62 beads (b) Siti has 3 × 62 = 186; 2 units = 124 more than Ravi

Q4 Take away the extra: 150 − 24 = 126; 126 ÷ 2 = 63. Ben 63, Ali 63 + 24 = 87. Check 63 + 87 = 150.

Q5 (a) 19 stickers (difference 38, give half; both have 66) (b) 38 is the whole difference. Giving 38 would leave Mei with 47 and Siti with 85: they just swap. Mei must give half of the difference.

Q6 (a) $27 (Ali $55, Ben $28) (b) $27 (c) The difference stays the same, because both boys got the same amount.

Sneak-peek puzzles Puzzle 1: 20 Puzzle 2: $14

Investigation (What stays the same?) Model answer and what to look for: Table (Mei, brother, difference, total): now 13, 5, 8, 18; 1: 14, 6, 8, 20; 2: 15, 7, 8, 22; 3: 16, 8, 8, 24; 4: 17, 9, 8, 26; 5: 18, 10, 8, 28. Same: the difference is always 8. Changes: both ages go up by 1 and the total goes up by 2 each year. Double: in 3 years (16 and 8). Mei is double when her brother's age equals the difference, 8. Triple:

never again. She was triple last year (12 and 4), and after that the ratio only gets smaller. Look for: using the table, noticing the difference is constant, and a bar-model reason for 'double' (her brother's bar is the same length as the difference). Extension: when did she last have 3 times his age?

Story break try-it (Bhaskara II, optional) 20 pearls. Half and a quarter make three quarters, so the 5 left is one quarter. One quarter is 5, so the whole is 4 x 5 = 20. Check: 10 + 5 + 5 = 20.

Go further (explore slide) Make 3 towers from 30 cubes, each 2 cubes taller than the last. How tall is each? (Then 3 taller.) Answer: 30 cubes: 8, 10, 12 (3 towers, take away 2 + 4 = 6, share 24 into 3). With 3 taller each time: 7, 10, 13 (30 - 9 = 21, 21 / 3 = 7).

Lesson 5: Remainder Riddles

Week 3 (first lesson) • Division • Workbook: Lesson 5

Objective	Divide with remainders, check that the remainder is smaller than the number we divide by, decide what the leftover means in a story, work backwards from groups and leftovers, and solve "what number am I?" riddles by making a list.
Thinking skills	Division with remainders, interpreting the remainder (one more group or leave it out), working backwards (groups × size + leftover), making a systematic list and testing each clue, making a table
Materials	25 counters per child (or beans or buttons), 6 small cups for groups, mini whiteboards and markers. For the sneak-peek puzzles: paper, a pencil and a few counters or small objects for each child.
Online set-up	Children use 25 small things at home (dried beans, buttons, LEGO bricks) and a few cups or bowls. Share the screen for the slides; children hold whiteboards up to the camera. Print the workbook before the lesson. Pairs work in one breakout room for 10 minutes, then return for the check slide.
Prepare	Solve the worksheet yourself first. Set out one materials box per pair. Open the slides for this lesson and check the answers in the speaker notes.

0–2 min Opening

SAY *"Welcome, Remainder Riddlers! When we share things out, there are often some left over. Today the leftovers are clues: they tell us how many buses to book, which number is hiding, and how many tables are in the room."*

DO Greet each child by name. Show the lesson title slide. Read "Today we will…" aloud.

Ask the whole group once: "What is still tricky from last time? Did anyone try the Investigation at home? Volunteers can share one thing." Pass is fine.

2–10 min Sneak-peek puzzles (2–5), warm-up game (5–8) and quick questions (8–10)

SAY *"Here are two sneak-peek puzzles. These are ideas we have not learnt yet, so guessing is welcome and so is "I do not know". Nobody is marked. Write your answer on your own paper. When I say "show me", hold it up to the camera for me."*

DO Show the "Sneak-peek puzzles" slide (minutes 2 to 5). Read puzzle 1 aloud and give about 1 minute: it is an access puzzle, so any child can try it with objects, a drawing or counting. Then puzzle 2, about 1 minute, the thinking puzzle. Look at the papers on camera quietly. Do not show the answers on the slide and do not read answers out. Say "thank you" to everyone in the same way.

Nobody is labelled. Never say who got it, who is on which page, or "fast group" or "top group". Do not single anyone out. Invite the whole group once Start here is done. It is an invitation, never a

ranking: every child can choose Think harder or the Investigation once page 1 and Q4 and Q5 of page 2 are done. We come back to puzzle 2 in talk time, for volunteers.

Sneak-peek puzzle	Answer (tutor only) and what to listen for
Puzzle 1 (access): Mei puts 13 sweets in bags of 4. Use counters or draw it. How many full bags can she make? How many sweets are left over?	3 full bags, 1 left over. Listen for: 3 bags of 4 use 12 sweets. 1 sweet is left.
Puzzle 2 (thinking): Ali says, "6 sweets are left over when I put the sweets in bags of 6." Draw 6 sweets in bags of 6. What is wrong?	6 sweets fill a bag, so 0 are left. Listen for: A leftover must be fewer than 6, otherwise another bag of 6 can be filled.

Choice during the lesson (groups of 8, online)

- Minutes 10 to 40 (explore, mini-lesson, brain break): everyone does the same. The explore slide has a Go further what-if from minute 10, so a pair that has finished its first task has real work at once. Do not say who is ready for it.
- Minute 40: everyone opens Start here page 1 and works in pairs. A line at the top says: “Do page 1, then Q4 and Q5 on page 2. After that you may choose: Think harder, Go further or the Investigation.” You never announce who is on which page. Invite the whole group, never one child: “When you have finished Start here you can choose Think harder, Go further or the Investigation.” Do not split a pair unless both children agree.
- The first-step help for the Investigation is printed on page 2 of Start here, so every child sees it. Pairs work in breakout rooms, so visit EVERY room by minute 47 (about a minute each, camera on, a parent nearby) and ask “how do you know?”, not “is it done?”. At about minute 52 check in again with two or three children, including children on Start here.
- Minute 67: everyone chooses what to do next: finish page 2, Think harder, Go further or the Investigation. The Investigation has its own slot, minutes 67 to 72, and its debrief opens talk time. Everyone is back together for talk time at minute 72. An Investigation that is not finished goes home; volunteers share one thing at the start of next lesson.
- Next lesson everything starts again from the same pages: nobody is on a track.

DO Warm-up game (minutes 5 to 8). Now the game, then the quick questions (minutes 8 to 10): Leftover Lucky Dip. The tutor calls a number of sweets and a group size (23 in 4s, 17 in 3s, 26 in 8s, 41 in 6s). Children show the groups on one hand and the leftover on the other.

SAY *“I will call out some sweets and a group size. Show me the number of groups on one hand and the leftover on the other.”*

“23 sweets in groups of 4. How many groups? How many left over? Which times-table fact helped you?”

DO After the game, ask the quick questions on the warm-up slides. Use thumbs or a whisper to partners before taking answers.

Ask	Expect

17 ÷ 5	3 r 2
30 ÷ 7	4 r 2
45 ÷ 6	7 r 3
50 ÷ 8	6 r 2
63 ÷ 9	7 r 0 (nothing left over)

10–25 min Explore (in pairs, with objects)

SAY *"We will put 25 counters into groups of 2, 3, 4, 5 and 6. Which group size do you think leaves nothing over? Write a guess."*

DO 1. Take 25 counters.

2. Put them in groups of 2, then 3, 4, 5 and 6.

3. Record the groups and the leftover in a table.

4. Ask: what do you notice about the leftovers?

SAY *"Talk to your partner while you work. I will come round and ask you questions."*

IF STUCK Stuck: make the groups with counters and push the leftovers to one side.

Use a times-table fact: 6 × 4 = 24, so 25 in groups of 4 is 6 groups and 1 left over.

Ready for more: the Go further what-if on the slide: try 49 counters. Which group sizes leave 1 over? Why?

DO Pairs work from minute 10. If a pair finishes the first task, they may take the Go further what-if on the slide (do not say who). At about minute 15 bring the group back together and take the three questions below, one at a time (minutes 15 to 25).

Ask	Expect
Which group size left nothing over?	Groups of 5: 25 is in the 5 times table (5 × 5).
Why did 2s, 3s, 4s and 6s all leave 1 over?	24 is in the 2, 3, 4 and 6 tables, and 25 is one more than 24.
In groups of 4, can 4 or more be left over?	No. If 4 were left, they would make another group. The leftover is always smaller than the group size.

25–35 min Mini-lesson (you teach, they join in)

DO Show the mini-lesson slides. Think aloud as you work. Say what you notice, what you test and how you check. Children copy how you think.

SAY *"38 sweets for 5 children. I count in 5s: 5 × 7 = 35, 5 × 8 = 40, too many. So each child gets 7."*

"I used 35, so 38 – 35 = 3 are left over. Is 3 smaller than 5? Yes. So 38 ÷ 5 = 7 r 3. Check: 7 × 5 + 3 = 38."

"50 people, vans of 8: 50 ÷ 8 = 6 r 2. Those 2 people still need a ride, so 7 vans. But 50 cakes in boxes of 8 fill only 6 boxes. Same division, different answer, so I read the question again."

Worked example on the slides	Answer
Share 38 sweets equally among 5 children. Count in 5s: 5 × 7 = 35, 5 × 8 = 40 is too many. 38 − 35 = 3 left over, and 3 is smaller than 5. Check: 7 × 5 + 3 = 38.	7 each, 3 left over: 38 ÷ 5 = 7 r 3.
50 people, vans of 8. How many vans?. 50 ÷ 8 = 6 r 2. The 2 people left over still need a ride. Compare: 50 cakes in boxes of 8 fill only 6 boxes.	7 vans (but only 6 full boxes of cakes).

SAY *"The key idea today is: read the leftover. The remainder is always smaller than the number we divide by. Start number = groups × size + leftover. Say it with me."*

DO Show the key idea slide at minutes 33 to 35. The "common mistakes" on that slide are for you, not the children. Watch for them during the worksheet:

- Leaving a remainder that is as big as, or bigger than, the number we divide by (60 ÷ 7 = 7 r 11).
- Giving the division answer when the story needs one more group (minibuses, boxes for ALL the eggs), or adding one when only full groups count (books she can buy).
- Stopping at the first number that fits one clue in a riddle, without testing the other clue or finding ALL the numbers.

35–40 min Brain break

DO Clap-and-stomp groups. The tutor calls "Clap 17 in 5s!" Everyone claps in groups of 5, then stomps once for each leftover clap. Then 23 in 4s and 20 in 6s.

Water. Toilet break for anyone who needs it. Keep it to five minutes.

DO Story break (optional, 2 minutes): after the movement, show the Story break slide (Sun Zi, China, lived more than 1,500 years ago (exact dates unknown)) and read the story aloud. Link to today: Today's "what number am I?" riddles use division with remainders, just like Sun Zi's riddle. Then offer the try-it as a question for everyone to think about: "undefined" Anyone may pass, nobody is called on by name and nothing is marked. The answer is in the notes of the slide and in the answer key below. Skip it if you are short of time: click straight on to the worksheet.

DO Accuracy note for you (do not read out): Dates and author of the Sunzi Suanjing are uncertain; the story says so.

40–72 min Worksheet (Start here pages, then your choice, with its own Investigation slot at 67–72)

DO Minute 40: give out the workbook. Read the title of Lesson 5 aloud. Say "Pairs first. Talk before you write."

Minutes 40 to 52: page 1 (Start here) in pairs; where the deck has a short teaching slide, it comes first. Pairs are in breakout rooms, so visit EVERY room by minute 47 (about a minute each; camera on, parent nearby), then check in again at about minute 52. Minutes 52 to 54: check page 1

together. Minutes 54 to 58: short teaching before page 2. Minutes 58 to 65: each child works alone on page 2. Minutes 65 to 67: check page 2 together. Minutes 67 to 72: everyone chooses Think harder, Go further or the Investigation (the Investigation has its own slide, minutes 67 to 72). A child who finishes early starts on one of them at once, and any child may choose them once page 1 and Q4 and Q5 of page 2 are done.

SAY *"Pairs first. Read each question aloud and talk about it before you write. I am not looking for speed. I am looking for good thinking."*

DO Circulate and ask:

- Ask: "Is your remainder smaller than the number you divided by?"
- Ask: "What happens to the leftover in this story?"
- Ask: "How do you know there are no more numbers in your list?"

IF STUCK Hint 1: "Which times table do you need? Count up in it."

Hint 2: "Groups × size + leftover gives the starting number."

Hint 3: "List the numbers that fit one clue. Then test each one with the other clue."

DO Anyone who wants more may choose Think harder first, then Go further, or the Investigation (its easy first step is on page 2 of Start here). Invite the whole group: "When you have finished Start here you can choose Think harder, Go further or the Investigation." Never announce who is on which page, and do not split a pair unless both children agree.

Think harder and Go further questions

Question	Answer
Workbook Think harder 1: Mei has 53 beads. She makes bracelets with 8 beads each. How many bracelets can she make? How many more beads does she need to make one more bracelet?	6 bracelets (53 ÷ 8 = 6 r 5). She needs 3 more beads (56 − 53).
Workbook Think harder 2: I am a number between 20 and 30. When I am divided by 4, the remainder is 3. When I am divided by 6, the remainder is 5. What number am I?	23 (remainder 3 when ÷ 4: 23 and 27; 23 ÷ 6 = 3 r 5, 27 ÷ 6 = 4 r 3)
Workbook Think harder 3: What is the biggest 2-digit number that leaves a remainder of 8 when it is divided by 9?	98 (9 × 10 + 8 = 98; 99 ÷ 9 has no remainder)
Workbook Think harder 4: A boat can carry 9 people at a time. 100 people want to cross the river. How many boatloads are needed?	12 boatloads (100 ÷ 9 = 11 r 1; the last person needs a boatload too)

Workbook Go further 5: Make up your own remainder riddle: "I am a number between ___ and ___. When I am divided by ___ the remainder is ___. When I am divided by ___ the remainder is ___." Check that only ONE number fits, by listing the numbers for the first clue and testing them. Swap with a partner.	Answers vary. Check that the number fits both clues and only ONE number does. Example: "Between 20 and 50; ÷ 5 leaves 2; ÷ 6 leaves 4." Only 22 fits (22 = 4 × 5 + 2 = 3 × 6 + 4). With the range 20 to 60, 52 would fit too.
Workbook Go further 6: Today is Monday. What day of the week will it be 100 days from today? What day of the week was it 100 days ago? Explain how you know.	100 days from today: Wednesday. 100 ÷ 7 = 14 remainder 2, so after 14 weeks it is Monday again, then 2 more days. 100 days ago: Saturday. Go back 14 weeks to a Monday, then 2 more days back: Sunday, then Saturday.
Workbook Go further 7: Mr Lim gives pens to his pupils. If each pupil gets 4 pens, 10 pens are left over. If each pupil gets 6 pens, he is 4 pens short. How many pupils are there? How many pens does he have?	7 pupils and 38 pens (giving 2 more pens each needs 10 + 4 = 14 pens, and 14 ÷ 2 = 7; 7 × 4 + 10 = 38, 7 × 6 = 42 = 38 + 4)

These are the questions on the Think harder and Go further pages in the workbook. More are in the separate Question Bank.

72–80 min Talk time (Investigation debrief first)

DO Bring the group back together (camera on). First 3 minutes (72 to 75): the Investigation debrief. Ask for volunteers to share what they noticed, then show the model answer (in the Investigation row at the end of this lesson's answer key and in the notes of the Investigation slide). An unfinished Investigation can be shared at the start of next lesson. Then the talk question. Ask for volunteers first. A pass is always fine, and so is sharing with a partner. Do not call a child by name cold, and keep no public tallies or rankings. Nobody is called on by name. Come back to sneak-peek puzzle 2: "Now we know the idea, how could it help?"

SAY "*Why can the remainder never be as big as the number we divide by?*"

"*75 ÷ 9 = 8 r 3. Why was the minibus answer 9, not 8?*"

Discussion question	Listen for
Why can the remainder never be as big as the number we divide by?	If the leftover were as big as the group size, we could make one more group, so the remainder is always smaller. The 3 pupils left over still need a seat, so we need one more minibus: 9.

80–90 min Exit ticket (80–86), look back (86–88) and closing (88–90)

DO Minutes 80 to 86. Give each child a slip of paper (or ask them to write on their journal page). No talking, no helping. Show the exit ticket slide. Children show the answer to you on camera in the open call (or keep it to share if they want); no photos and no private messages. Never mark or read answers out in front of the group.

Exit ticket	Answer
29 children go on a ride. Each car holds 4 children. How many cars are needed? Write the division and explain what the leftover means.	29 ÷ 4 = 7 r 1. 8 cars are needed: the 1 child left over still needs a car.

Look back (minutes 86 to 88, two minutes)

SAY *"Turn to the last page of today's lesson and find Look back. Tick or write one thing for each line: something you did well, the hardest part, and what helped you. There are no right answers, and you may pass on any line."*

"Would anyone like to share something they did well, or what helped? Thank you. When you got stuck and tried something else, that is exactly what mathematicians do."

DO Give a full 2 minutes. Never collect the page, mark it or read it out in front of the group: children keep it. You cannot see the paper. Tell the WHOLE group: "If you ticked still stuck, say check-in please in the open class, or ask your grown-up to tell me." Then talk it through with that child in the open session, with a parent nearby. No hand signals that single a child out, never a private message. Open the next lesson with the whole group: "What is still tricky from last time?" Invite volunteers only. Praise the strategy ("you drew it and that got you unstuck"), not the speed or the answer.

SAY *"Today you read the leftovers like clues. Inquirers ask what an answer means, not just what it is."*

DO Minutes 88 to 90. Stamp the inquiry journal when a child has shown you their exit ticket. Mark S, D or H in the progress tracker at the back of this guide (it is for you only). If a child is marked D or H, plan two minutes of quiet catch-up at the start of the next lesson.

Extension for children who want more

Find the smallest number bigger than 1 that leaves a remainder of 1 when it is divided by 2, 3, 4, 5 and 6. (61: it is one more than 60, which is in all five tables.)

Answer key (workbook)

Q1 (a) 7 each, 5 left over (6 × 7 = 42, 47 − 42 = 5). (b) 1 more (48 = 6 × 8). (c) The remainder 11 is bigger than 7, so another group of 7 fits: 60 ÷ 7 = 8 r 4 (7 × 8 = 56, 60 − 56 = 4).

Q2 (a) 75 ÷ 9 = 8 r 3, so 9 minibuses (the 3 pupils left over need a bus too). (b) 60 ÷ 7 = 8 r 4, so 8 books ($4 is not enough for another). (c) 87 ÷ 6 = 14 r 3: 14 full boxes. (d) 15 boxes to hold all the eggs.

Q3 (a) 12 × 8 + 5 = 96 + 5 = 101 marbles. (b) 101 ÷ 9 = 11 r 2: 11 full bags, 2 marbles left over (9 × 11 = 99).

Q4 (a) Remainder 3 when ÷ 5: 43, 48, 53, 58. Remainders when ÷ 4: 3, 0, 1, 2. I am 53. (b) 11, 23, 35 and 47 (remainder 5 when ÷ 6: 11, 17, 23, 29, 35, 41, 47; only 11, 23, 35, 47 leave 3 when ÷ 4). They go up in 12s.

Q5 (a) 0, 1, 2, 3, 4, 5, 6, 7 (the remainder must be smaller than 8). (b) 63 (biggest remainder 7; 56 + 7 = 63; 64 + 7 = 71 is too big).

Q6 (a) 5 tables and 35 children (5 × 6 + 5 = 35 and 5 × 7 = 35). In the table the gap shrinks by 1 each row: 1 table 11 and 7, 2 tables 17 and 14, 3 tables 23 and 21, 4 tables 29 and 28, 5 tables 35 and 35. (b) 5 × 8 = 40 seats, 40 − 35 = 5 empty seats.

Sneak-peek puzzles Puzzle 1: 3 full bags, 1 left over Puzzle 2: 6 sweets fill a bag, so 0 are left

Investigation (Is it always true? Three in a row) Model answer and what to look for: Always true: among 3 numbers in a row there is exactly one in the 3 times table. Remainders when dividing by 3 go 1, 2, 0, 1, 2, 0, … so in any 3 in a row each remainder 0, 1, 2 appears once, and remainder 0 means it is in the table. With 4 in a row and the 4 times table it is also exactly one (remainders 0, 1, 2, 3 repeat). With only 2 in a row and the 3 times table it is NOT always true (4 and 5 have none). In general, n numbers in a row always have exactly one in the n times table. Look for: several tests including a big number, the remainder cycle as the reason, and a counterexample for the 2 in a row case.

Story break try-it (Sun Zi, optional) 23 / 3 = 7 r 2, 23 / 5 = 4 r 3, 23 / 7 = 3 r 2, so 23 works. The second riddle: 17 (17 / 4 = 4 r 1 and 17 / 3 = 5 r 2). Of 13 and 17, only 17 fits.

Go further (explore slide) 49 counters: which group sizes from 2 to 8 leave 1 over? Why? Answer: 49 = 48 + 1. Sizes 2, 3, 4, 6 and 8 leave 1 over (48 is in all those tables). Size 5 leaves 4 and size 7 leaves 0 (49 = 7 x 7).

Lesson 6: Fraction Walls

Week 3 (second lesson) • Fractions • Workbook: Lesson 6

Objective	Use a fraction wall to find equivalent fractions, compare and order fractions by making the parts the same size, and add and subtract like and related fractions within one whole.
Thinking skills	Drawing a model (fraction wall, bar), equivalent fractions (multiply top and bottom by the same number), comparing with the same bottom number or the same top number, adding related fractions, finding all fractions between two fractions
Materials	The printed fraction strip page (7 strips per child, all the same length, 24 cm), coloured pencils, scissors, mini whiteboards. For the sneak-peek puzzles: paper, a pencil and a few counters or small objects for each child.
Online set-up	Send the printed fraction strip page instead of asking children to measure and cut strips. Marks every 12, 8, 6, 4, 3 and 2 cm give halves, thirds, quarters, sixths, eighths and twelfths. Share the screen for the slides. Print the workbook before the lesson. Pairs work in one breakout room for 10 minutes, then return for the check slide.
Prepare	Solve the worksheet yourself first. Set out one materials box per pair. Open the slides for this lesson and check the answers in the speaker notes.

0–2 min Opening

SAY *"Welcome, wall builders! Today we build a fraction wall and use it to spot fractions that look different but are the same amount, to decide which fraction is bigger, and to add fractions without falling into the bottom-number trap."*

DO Greet each child by name. Show the lesson title slide. Read "Today we will…" aloud.

Ask the whole group once: "What is still tricky from last time? Did anyone try the Investigation at home? Volunteers can share one thing." Pass is fine.

2–10 min Sneak-peek puzzles (2–5), warm-up game (5–8) and quick questions (8–10)

SAY *"Here are two sneak-peek puzzles. These are ideas we have not learnt yet, so guessing is welcome and so is "I do not know". Nobody is marked. Write your answer on your own paper. When I say "show me", hold it up to the camera for me."*

DO Show the "Sneak-peek puzzles" slide (minutes 2 to 5). Read puzzle 1 aloud and give about 1 minute: it is an access puzzle, so any child can try it with objects, a drawing or counting. Then puzzle 2, about 1 minute, the thinking puzzle. Look at the papers on camera quietly. Do not show the answers on the slide and do not read answers out. Say "thank you" to everyone in the same way.

Nobody is labelled. Never say who got it, who is on which page, or "fast group" or "top group". Do not single anyone out. Invite the whole group once Start here is done. It is an invitation, never a ranking: every child can choose Think harder or the Investigation once page 1 and Q4 and Q5 of page 2 are done. We come back to puzzle 2 in talk time, for volunteers.

Sneak-peek puzzle	Answer (tutor only) and what to listen for
Puzzle 1 (access): Draw a rectangle. Cut it into 4 equal parts and shade 1 part. What fraction did you shade?	1/4. Listen for: 4 equal parts, 1 shaded: one quarter.
Puzzle 2 (thinking): Mei says, "1/4 of a pizza is bigger than 1/3 of the same pizza, because 4 is bigger than 3." Is she right? Draw it.	No: 1/3 is bigger. Listen for: More equal parts means each part is smaller. Quarters are smaller than thirds.

Choice during the lesson (groups of 8, online)

- Minutes 10 to 40 (explore, mini-lesson, brain break): everyone does the same. The explore slide has a Go further what-if from minute 10, so a pair that has finished its first task has real work at once. Do not say who is ready for it.
- Minute 40: everyone opens Start here page 1 and works in pairs. A line at the top says: “Do page 1, then Q4 and Q5 on page 2. After that you may choose: Think harder, Go further or the Investigation.” You never announce who is on which page. Invite the whole group, never one child: “When you have finished Start here you can choose Think harder, Go further or the Investigation.” Do not split a pair unless both children agree.
- The first-step help for the Investigation is printed on page 2 of Start here, so every child sees it. Pairs work in breakout rooms, so visit EVERY room by minute 47 (about a minute each, camera on, a parent nearby) and ask “how do you know?”, not “is it done?”. At about minute 52 check in again with two or three children, including children on Start here.
- Minute 67: everyone chooses what to do next: finish page 2, Think harder, Go further or the Investigation. The Investigation has its own slot, minutes 67 to 72, and its debrief opens talk time. Everyone is back together for talk time at minute 72. An Investigation that is not finished goes home; volunteers share one thing at the start of next lesson.
- Next lesson everything starts again from the same pages: nobody is on a track.

DO Warm-up game (minutes 5 to 8). Now the game, then the quick questions (minutes 8 to 10): Fraction Flash. Four shapes cut into equal parts appear on screen (eighths of a circle, sixths of a rectangle, sixths of a hexagon, tenths of a bar). Children write the fraction shaded, then click to check.

SAY *“Look at the shape. How many equal parts? How many are shaded? Write the fraction on your whiteboard.”*

“How do you know the parts are equal? What would go wrong if they were not?”

DO After the game, ask the quick questions on the warm-up slides. Use thumbs or a whisper to partners before taking answers.

Ask	Expect
How many quarters make one whole?	4
Which is bigger: 1/3 or 1/5?	1/3

1/2 = ?/10	5
2/9 + 4/9	6/9
1 – 2/5	3/5

10–25 min Explore (in pairs, with objects)

SAY *"Which is bigger: one sixth of a strip or one eighth of the same strip? Write a guess before we start."*

DO 1. Take 7 paper strips, all the same length.

2. Keep one whole. Use the printed strips (or fold or mark paper strips) for 2, 4, 8, 3, 6 and 12 equal parts.

3. Label every part and stack the strips into a wall.

4. Ask: which pieces line up exactly?

SAY *"Talk to your partner while you work. I will come round and ask you questions."*

IF STUCK Stuck: look at the printed strips, or fold a paper strip in half, then in half again, then again. How many parts now?

For thirds: roll the strip into an S shape until the three parts are the same, or mark every 8 cm.

Ready for more: find three different pieces that together cover exactly one half.

DO Pairs work from minute 10. If a pair finishes the first task, they may take the Go further what-if on the slide (do not say who). At about minute 15 bring the group back together and take the three questions below, one at a time (minutes 15 to 25).

Ask	Expect
How many 1/8 pieces fit exactly on 1/2?	4 (1/2 = 4/8).
Which is bigger, 1/6 or 1/8? Why?	1/6: the whole is cut into fewer parts, so each part is bigger.
Can eighths cover 1/3 exactly?	No. 2 eighths are too small and 3 eighths are too big. Sixths and twelfths can (2/6, 4/12).

25–35 min Mini-lesson (you teach, they join in)

DO Show the mini-lesson slides. Think aloud as you work. Say what you notice, what you test and how you check. Children copy how you think.

SAY *"1/3 on the wall lines up with 2 sixths and with 4 twelfths. Same amount, different pieces: 1/3 = 2/6 = 4/12."*

"From 1/3 to 2/6, the top and the bottom were both doubled. From 1/3 to 4/12, both times 4. So 2/5 = 4/10 without a wall."

"5/6 or 3/4? The parts are different sizes, so I change both to twelfths: 10/12 and 9/12. Now the parts match: 5/6 is bigger."

Worked example on the slides	Answer
Equal fractions on the wall: 1/3. 1/3 lines up with 2 sixths and with 4 twelfths. Top and bottom are multiplied by the same number (× 2, × 4). So 2/5 = 4/10 without a wall.	1/3 = 2/6 = 4/12.
Which is bigger: 5/6 or 3/4?. Different-sized parts, so make them the same. 5/6 = 10/12 and 3/4 = 9/12. Now compare the top numbers.	5/6 is bigger (10/12 > 9/12).

SAY *"The key idea today is: make the parts the same size. To compare or add fractions, the parts must be the same size. Use the wall, or change one fraction to an equal fraction with the same bottom number. Say it with me."*

DO Show the key idea slide at minutes 33 to 35. The "common mistakes" on that slide are for you, not the children. Watch for them during the worksheet:

- Adding the bottom numbers as well as the top numbers (1/3 + 1/3 = 2/6, or 1/4 + 1/8 = 2/12).
- Thinking 1/8 is bigger than 1/6 because 8 is bigger than 6.
- Comparing or adding fractions with different-sized parts without changing one first (1/4 + 3/8 = 4/8).

35–40 min Brain break

DO Fraction turns. The tutor calls a turn: "a quarter turn!", "a half turn!", "three quarters of a turn!", "two quarters!" Children turn on the spot and freeze. Which calls make the same turn?

Water. Toilet break for anyone who needs it. Keep it to five minutes.

DO Story break (optional, 2 minutes): after the movement, show the Story break slide (Abu Bakr al-Hassar, Morocco / the Islamic West (sources differ between the Maghreb and Spain), 12th century (exact dates unknown)) and read the story aloud. Link to today: Today you read and compare fractions, and the bar between the top and bottom numbers is the one al-Hassar described. Then offer the try-it as a question for everyone to think about: "undefined" Anyone may pass, nobody is called on by name and nothing is marked. The answer is in the notes of the slide and in the answer key below. Skip it if you are short of time: click straight on to the worksheet.

DO Accuracy note for you (do not read out): His dates and place are uncertain, and the story says so. The claim that Fibonacci's use of the bar came after al-Hassar is stated by MAA and Muslim Heritage. The sources do not say whether Fibonacci learned the bar from al-Hassar.

40–72 min Worksheet (Start here pages, then your choice, with its own Investigation slot at 67–72)

DO Minute 40: give out the workbook. Read the title of Lesson 6 aloud. Say "Pairs first. Talk before you write."

Minutes 40 to 52: page 1 (Start here) in pairs; where the deck has a short teaching slide, it comes first. Pairs are in breakout rooms, so visit EVERY room by minute 47 (about a minute each; camera on, parent nearby), then check in again at about minute 52. Minutes 52 to 54: check page 1 together. Minutes 54 to 58: short teaching before page 2. Minutes 58 to 65: each child works alone

on page 2. Minutes 65 to 67: check page 2 together. Minutes 67 to 72: everyone chooses Think harder, Go further or the Investigation (the Investigation has its own slide, minutes 67 to 72). A child who finishes early starts on one of them at once, and any child may choose them once page 1 and Q4 and Q5 of page 2 are done.

SAY *"Pairs first. Read each question aloud and talk about it before you write. I am not looking for speed. I am looking for good thinking."*

DO Circulate and ask:

- Ask: "Are the parts the same size? How do you know?"
- Ask: "Which fraction did you change, and what did you multiply by?"
- Ask: "You added the bottom numbers. Show me on the bar: did the pieces change size?"

IF STUCK Hint 1: "Find both fractions on the wall. Which one reaches further?"

Hint 2: "Change one fraction so both have the same bottom number."

Hint 3: "Draw a bar and cut it into the smaller pieces."

DO Anyone who wants more may choose Think harder first, then Go further, or the Investigation (its easy first step is on page 2 of Start here). Invite the whole group: "When you have finished Start here you can choose Think harder, Go further or the Investigation." Never announce who is on which page, and do not split a pair unless both children agree.

Think harder and Go further questions

Question	Answer
Workbook Think harder 1: Find ALL the fractions with bottom number 12 that are bigger than 1/2 but smaller than 3/4. Hint: look at the twelfths row of your wall. Which one is equal to 2/3?	7/12 and 8/12 (1/2 = 6/12 and 3/4 = 9/12; the end fractions are not between). 8/12 = 2/3.
Workbook Think harder 2: Which is bigger, 3/5 or 7/10? Show how you know.	7/10 (3/5 = 6/10, and 7/10 is bigger than 6/10)
Workbook Think harder 3: Siti reads 1/4 of a book on Monday and 1/2 of the book on Tuesday. What fraction of the book has she read? What fraction is left?	She has read 3/4 (1/4 + 2/4). 1/4 is left.
Workbook Think harder 4: Put these in order, smallest first: 2/3, 5/6, 7/12.	7/12, 2/3, 5/6 (7/12, 8/12, 10/12)
Workbook Go further 5: Make up your own "What fraction am I?" riddle, like: "I am equal to 3/4. My top and bottom numbers add up to ___." Choose the number so that exactly ONE fraction fits. Check it, then swap with a partner. Is your riddle always going to work for any number you choose? Explain.	Answers vary. Check that the fraction is equal to the given one and that the clue fits only one fraction. Example: "I am equal to 3/4. My top and bottom numbers add up to 21." Fractions equal to 3/4 are 3/4, 6/8, 9/12, 12/16 ..., with sums 7, 14, 21, 28 ..., so only 9/12 fits. The riddle only works if the number is in the 7 times table (7, 14, 21, 28 ...).

Workbook Go further 6: Fill one whole bar using exactly 5 pieces. You may use 1/2, 1/4 and 1/8 pieces, as many of each as you like. Find all the ways.	2 ways: 1/2 + 1/8 + 1/8 + 1/8 + 1/8, and 1/4 + 1/4 + 1/4 + 1/8 + 1/8.
Workbook Go further 7: Find a fraction that is bigger than 2/3 but smaller than 3/4. Hint: the fraction wall has no row that helps, so cut every part into smaller equal parts. Can you find another one?	17/24 (2/3 = 16/24 and 3/4 = 18/24). Another: 25/36 (2/3 = 24/36, 3/4 = 27/36). Many answers are possible: cut the parts smaller and smaller.

These are the questions on the Think harder and Go further pages in the workbook. More are in the separate Question Bank.

72–80 min Talk time (Investigation debrief first)

DO Bring the group back together (camera on). First 3 minutes (72 to 75): the Investigation debrief. Ask for volunteers to share what they noticed, then show the model answer (in the Investigation row at the end of this lesson's answer key and in the notes of the Investigation slide). An unfinished Investigation can be shared at the start of next lesson. Then the talk question. Ask for volunteers first. A pass is always fine, and so is sharing with a partner. Do not call a child by name cold, and keep no public tallies or rankings. Nobody is called on by name. Come back to sneak-peek puzzle 2: "Now we know the idea, how could it help?"

SAY *"Why is 1/8 smaller than 1/6, even though 8 is bigger than 6?"*

"Why is 1/3 + 1/3 not 2/6?"

Discussion question	Listen for
Why is 1/8 smaller than 1/6, even though 8 is bigger than 6?	The same whole is cut into more pieces, so each eighth is smaller than each sixth. 1/3 + 1/3 is two thirds: the pieces are still thirds, so we add only the top numbers (2/6 is just 1/3).

80–90 min Exit ticket (80–86), look back (86–88) and closing (88–90)

DO Minutes 80 to 86. Give each child a slip of paper (or ask them to write on their journal page). No talking, no helping. Show the exit ticket slide. Children show the answer to you on camera in the open call (or keep it to share if they want); no photos and no private messages. Never mark or read answers out in front of the group.

Exit ticket	Answer
Ali says 1/4 + 1/8 = 2/12. Is he right? Write the correct answer and show how you know.	No. 1/4 = 2/8, so 1/4 + 1/8 = 2/8 + 1/8 = 3/8. We add the parts, not the bottom numbers.

Look back (minutes 86 to 88, two minutes)

SAY *"Turn to the last page of today's lesson and find Look back. Tick or write one thing for each line: something you did well, the hardest part, and what helped you. There are no right answers, and you may pass on any line."*

"Would anyone like to share something they did well, or what helped? Thank you. When you got stuck and tried something else, that is exactly what mathematicians do."

DO Give a full 2 minutes. Never collect the page, mark it or read it out in front of the group: children keep it. You cannot see the paper. Tell the WHOLE group: "If you ticked still stuck, say check-in please in the open class, or ask your grown-up to tell me." Then talk it through with that child in the open session, with a parent nearby. No hand signals that single a child out, never a private message. Open the next lesson with the whole group: "What is still tricky from last time?" Invite volunteers only. Praise the strategy ("you drew it and that got you unstuck"), not the speed or the answer.

SAY *"Today you made the parts the same size before you compared or added. Inquirers check that a comparison is fair."*

DO Minutes 88 to 90. Stamp the inquiry journal when a child has shown you their exit ticket. Mark S, D or H in the progress tracker at the back of this guide (it is for you only). If a child is marked D or H, plan two minutes of quiet catch-up at the start of the next lesson.

Extension for children who want more

How many different ways can you fill one whole bar using 1/2, 1/4 and 1/8 pieces, as many of each as you like? (9 ways. Work in order: two halves; one half with the other half made of quarters and eighths in 3 ways; no halves, with 4, 3, 2, 1 or 0 quarters filled up with eighths: 5 ways.)

Answer key (workbook)

Q1 (a) 2, 3, 4, 6: 1/2 = 2/4 = 3/6 = 4/8 = 6/12. (b) 4, 8: 2/3 = 4/6 = 8/12. (c) 6, 9: 3/4 = 6/8 = 9/12. (d) The bottom number is always double the top number (the top and bottom are multiplied by the same number).

Q2 (a) > (b) < (c) = (d) > (3/4 = 6/8) (e) < (same top number; sevenths are bigger than ninths) (f) < (2/3 = 8/12, 3/4 = 9/12) (g) 1/3, 5/12, 1/2, 3/4 (4/12, 5/12, 6/12, 9/12).

Q3 (a) A, B and D (A: 1 of 4 equal parts; B: 2/8 = 1/4; D: 3/12 = 1/4). (b) C: 2/6 (= 1/3) shaded. E: 3/8 shaded.

Q4 (a) 5/7 (b) 2/8 + 3/8 = 5/8 (c) 4/12 + 5/12 = 9/12 (= 3/4) (d) 5/6 − 2/6 = 3/6 (= 1/2) (e) 10/10 − 3/10 = 7/10 (f) 5/10 + 3/10 = 8/10 (= 4/5).

Q5 (a) 1/6 + 4/6 = 5/6 of the fence. (b) 1/6 is not painted. (c) 4/6 − 1/6 = 3/6 = 1/2 of the fence more in the afternoon.

Q6 1/4 and 6/8; 2/3 and 4/12; 3/8 and 5/8; 5/12 and 7/12; 1/6 and 5/6.

Think harder 1 (a) 7/12 and 8/12 (1/2 = 6/12 and 3/4 = 9/12; the end fractions are not between). (b) 8/12 = 2/3.

Sneak-peek puzzles Puzzle 1: 1/4 Puzzle 2: No: 1/3 is bigger

Investigation (Find all the ways) Model answer and what to look for: 3/4 is the same as 6/8. The 6 ways (halves, quarters, eighths): 1 half + 1 quarter; 1 half + 2 eighths; 3 quarters; 2 quarters + 2 eighths; 1 quarter + 4 eighths; 6 eighths. Systematic: fix the number of halves (1 or 0), then the quarters, then fill with eighths. For 1 whole there are 9 ways, so more. Look for: a list in order, an organised method for 'all', and a reason that more room means more ways. Using the fraction wall is expected.

Story break try-it (Abu Bakr al-Hassar, optional) 2/3 = 8/12 and 3/4 = 9/12, so 3/4 is bigger.

Go further (explore slide) Find 3 DIFFERENT pieces that make exactly 1/2. Is there another way? Answer: Only one way with three different pieces: 1/4 + 1/6 + 1/12 (3/12 + 2/12 + 1/12 = 6/12). Other ways repeat a piece, e.g. 1/4 + 1/8 + 1/8.

Lesson 7: Area Architects

Week 4 (first lesson) • Area and Perimeter • Workbook: Lesson 7

Objective	Find the area and perimeter of figures made of squares and rectangles, find every rectangle with a given perimeter by listing in order, and use “change the shape” (push the steps out; cut up or fill in) to find perimeters and areas quickly.
Thinking skills	Systematic listing, changing the shape, spatial reasoning, explaining
Materials	12 square tiles (or 12 squares cut from 1 cm grid paper) per child, 1 cm grid paper, mini whiteboards, a ruler. For the sneak-peek puzzles: paper, a pencil and a few counters or small objects for each child.
Online set-up	Children cut 12 squares from squared paper (or draw on squared paper) and hold up their rectangles to the camera; the tutor shares a grid slide to draw on.
Prepare	Solve the worksheet yourself first. Set out one materials box per pair. Open the slides for this lesson and check the answers in the speaker notes.

0–2 min Opening

SAY *“Today you are architects. Architects plan rooms, gardens and fences, so they need two numbers: how much space is inside (the area) and how long the edge is (the perimeter).”*

“Here is a puzzle to keep in your head: can two gardens have the same fence but different amounts of grass? Let us find out.”

DO Greet each child by name. Show the lesson title slide. Read “Today we will…” aloud.

Ask the whole group once: “What is still tricky from last time? Did anyone try the Investigation at home? Volunteers can share one thing.” Pass is fine.

2–10 min Sneak-peek puzzles (2–5), warm-up game (5–8) and quick questions (8–10)

SAY *“Here are two sneak-peek puzzles. These are ideas we have not learnt yet, so guessing is welcome and so is “I do not know”. Nobody is marked. Write your answer on your own paper. When I say “show me”, hold it up to the camera for me.”*

DO Show the “Sneak-peek puzzles” slide (minutes 2 to 5). Read puzzle 1 aloud and give about 1 minute: it is an access puzzle, so any child can try it with objects, a drawing or counting. Then puzzle 2, about 1 minute, the thinking puzzle. Look at the papers on camera quietly. Do not show the answers on the slide and do not read answers out. Say “thank you” to everyone in the same way.

Nobody is labelled. Never say who got it, who is on which page, or “fast group” or “top group”. Do not single anyone out. Invite the whole group once Start here is done. It is an invitation, never a ranking: every child can choose Think harder or the Investigation once page 1 and Q4 and Q5 of page 2 are done. We come back to puzzle 2 in talk time, for volunteers.

Sneak-peek puzzle	Answer (tutor only) and what to listen for
Puzzle 1 (access): Draw a rectangle 3 cm long and 2 cm wide on squared paper (or use 6 tiles). How many squares does it cover? How far is it all the way round the edge?	6 squares; 10 cm round the edge. Listen for: 6 squares cover it. Walk round the edge: 3 + 2 + 3 + 2 = 10.
Puzzle 2 (thinking): Two different rectangles both cover 12 squares. Can the distance all the way round the edge be different? Show two examples.	Yes, e.g. 26 cm and 16 cm. Listen for: 1 by 12 is 26 cm round; 2 by 6 is 16 cm round; 3 by 4 is 14 cm round.

Choice during the lesson (groups of 8, online)

- Minutes 10 to 40 (explore, mini-lesson, brain break): everyone does the same. The explore slide has a Go further what-if from minute 10, so a pair that has finished its first task has real work at once. Do not say who is ready for it.
- Minute 40: everyone opens Start here page 1 and works in pairs. A line at the top says: “Do page 1, then Q4 and Q5 on page 2. After that you may choose: Think harder, Go further or the Investigation.” You never announce who is on which page. Invite the whole group, never one child: “When you have finished Start here you can choose Think harder, Go further or the Investigation.” Do not split a pair unless both children agree.
- The first-step help for the Investigation is printed on page 2 of Start here, so every child sees it. Pairs work in breakout rooms, so visit EVERY room by minute 47 (about a minute each, camera on, a parent nearby) and ask “how do you know?”, not “is it done?”. At about minute 52 check in again with two or three children, including children on Start here.
- Minute 67: everyone chooses what to do next: finish page 2, Think harder, Go further or the Investigation. The Investigation has its own slot, minutes 67 to 72, and its debrief opens talk time. Everyone is back together for talk time at minute 72. An Investigation that is not finished goes home; volunteers share one thing at the start of next lesson.
- Next lesson everything starts again from the same pages: nobody is on a track.

DO Warm-up game (minutes 5 to 8). Now the game, then the quick questions (minutes 8 to 10): Area or perimeter? Trace a finger round the edge for perimeter, wipe a flat palm for area: fence a garden, tile a floor, ribbon round a card, paint a wall, frame a photo, cover a book.

SAY *“When I read a job, show me with your hand. If the job goes round the edge, trace a finger round the edge in the air: perimeter. If the job covers the inside, wipe your flat palm: area.”*

“Ready? Job 1: put a fence round a garden.”

DO After the game, ask the quick questions on the warm-up slides. Use thumbs or a whisper to partners before taking answers.

Ask	Expect
8 × 7 = ?	56
A square has sides of 6 cm. What is its perimeter?	24 cm

A rectangle is 8 cm by 3 cm. What is its area?	24 cm²
A square has an area of 49 cm². How long is each side?	7 cm

10–25 min Explore (in pairs, with objects)

SAY *"You have 12 squares. You must use all of them to make a rectangle. Will every rectangle you make have the same perimeter? Thumbs up for yes, down for no. Write your guess on your whiteboard."*

DO 1. Take 12 squares.

2. Make a rectangle using ALL 12 squares.

3. Record its length, breadth and perimeter in a table.

4. Make every different rectangle. Which has the longest perimeter?

SAY *"Talk to your partner while you work. I will come round and ask you questions."*

IF STUCK Stuck: put 2 squares in each row. How many rows do you need for 12?

Not sure how to find the perimeter: run your finger round the edge and count every side of a square you pass.

Ready for more: can you make a rectangle with 12 squares that has a perimeter of 20 cm? (No: the perimeters are 26, 16 and 14 cm.)

DO Pairs work from minute 10. If a pair finishes the first task, they may take the Go further what-if on the slide (do not say who). At about minute 15 bring the group back together and take the three questions below, one at a time (minutes 15 to 25).

Ask	Expect
How many different rectangles did you make?	3: 12 by 1, 6 by 2 and 4 by 3 (turning one round does not make a new one).
Do they all have the same perimeter?	No: 26 cm, 16 cm and 14 cm. Same area, different perimeters.
Which has the longest perimeter? Why?	The long thin one (12 by 1): fewer squares touch, so more edges are on the outside.

25–35 min Mini-lesson (you teach, they join in)

DO Show the mini-lesson slides. Think aloud as you work. Say what you notice, what you test and how you check. Children copy how you think.

SAY *"Ravi has 20 m of fence. Half of it goes along one length and one breadth, so length + breadth = 10. I list in order: breadth 1, length 9; breadth 2, length 8; and so on. I stop at 5 and 5, because after that the rectangles just turn round."*

"This staircase has short steps with no numbers. I do not need them! I push the top pieces up and the side pieces out. Now it is a rectangle with the same perimeter: 8 + 6 + 8 + 6 = 28 cm."

"For the area of an L-shape I cut it into two rectangles. Then I check another way: I fill in the missing corner and take it away. Both ways give 42 cm², so I trust my answer."

Worked example on the slides	Answer
Same fence, different gardens. Ravi has 20 m of fence for a rectangular garden with whole-metre sides. Length + breadth = 20 ÷ 2 = 10. List in order, breadth 1 first. Which garden has the biggest area?	5 gardens. 5 m by 5 m has the biggest area: 25 m².
Push the steps out. A staircase shape is 8 cm wide and 6 cm tall. The steps have no lengths marked. Slide the pieces facing up to the top: together they make 8 cm. Slide the pieces facing right: together they make 6 cm. What is the perimeter?	8 + 6 + 8 + 6 = 28 cm

SAY *"The key idea today is: list in order, change the shape. Perimeter: push the steps out (but add the walls of a dent). Area: cut it into rectangles, or fill it in and take away. Say it with me."*

DO Show the key idea slide at minutes 34 to 35. The "common mistakes" on that slide are for you, not the children. Watch for them during the worksheet:

- Mixing up area and perimeter, or writing cm for an area (it is cm²).
- Counting the squares along the edge instead of the edges (for perimeter, count the sides of the squares, not the squares).
- Pushing out the sides of a shape with a dent: the two walls of a dent must be added on.

35–40 min Brain break

DO Perimeter march: march on the spot and trace a big square in the air, counting its 4 sides. Then stretch tall and thin like a 9 by 1 garden, then wide like a 5 by 5 garden.

Water. Toilet break for anyone who needs it. Keep it to five minutes.

DO Story break (optional, 2 minutes): after the movement, show the Story break slide (Karen Uhlenbeck, USA (born in Cleveland, Ohio), born 1942) and read the story aloud. Link to today: Today you find area and perimeter, and she studied shapes that use the smallest area for the edge that holds them. Then offer the try-it as a question for everyone to think about: "undefined" Anyone may pass, nobody is called on by name and nothing is marked. The answer is in the notes of the slide and in the answer key below. Skip it if you are short of time: click straight on to the worksheet.

DO Accuracy note for you (do not read out): The story uses the soap film only as a picture of 'least area'; the sources say a soap film settles into a minimal surface.

40–72 min Worksheet (Start here pages, then your choice, with its own Investigation slot at 67–72)

DO Minute 40: give out the workbook. Read the title of Lesson 7 aloud. Say “Pairs first. Talk before you write.”

Minutes 40 to 52: page 1 (Start here) in pairs; where the deck has a short teaching slide, it comes first. Pairs are in breakout rooms, so visit EVERY room by minute 47 (about a minute each; camera on, parent nearby), then check in again at about minute 52. Minutes 52 to 54: check page 1 together. Minutes 54 to 56: short teaching before page 2. Minutes 56 to 65: each child works alone on page 2. Minutes 65 to 67: check page 2 together. Minutes 67 to 72: everyone chooses Think harder, Go further or the Investigation (the Investigation has its own slide, minutes 67 to 72). A child who finishes early starts on one of them at once, and any child may choose them once page 1 and Q4 and Q5 of page 2 are done.

SAY *“Pairs first. Read each question aloud and talk about it before you write. I am not looking for speed. I am looking for good thinking.”*

DO Circulate and ask:

- Ask: “Is that the area or the perimeter? Which unit goes with it?”
- Ask: “How do you know you have found every rectangle?”
- Ask: “Can you push the steps out? Is there a dent?”

IF STUCK Hint 1: “Draw it on the grid and count.”

Hint 2: “Half the perimeter is one length plus one breadth.”

Hint 3: “Cut the shape into rectangles. Find each area, then add.”

DO Anyone who wants more may choose Think harder first, then Go further, or the Investigation (its easy first step is on page 2 of Start here). Invite the whole group: “When you have finished Start here you can choose Think harder, Go further or the Investigation.” Never announce who is on which page, and do not split a pair unless both children agree.

Think harder and Go further questions

Question	Answer
Workbook Think harder 1: Find all the rectangles with whole-cm sides and an area of 24 cm². Which one has the shortest perimeter?	24 by 1 (50 cm), 12 by 2 (28 cm), 8 by 3 (22 cm), 6 by 4 (20 cm). The 6 cm by 4 cm rectangle has the shortest perimeter.
Workbook Think harder 2: A rectangle is 12 cm long. Its perimeter is 30 cm. Find its area.	Length + breadth = 15, so the breadth is 3 cm. Area 12 × 3 = 36 cm².
Workbook Think harder 3: A rectangle 8 cm by 2 cm and a square have the same perimeter. Which has the bigger area, and by how much?	Perimeter 20 cm, so the square has sides of 5 cm. Square 25 cm², rectangle 16 cm². The square, by 9 cm².
Workbook Think harder 4: A square with sides of 6 cm is cut into two rectangles of the same size. What is the perimeter of each rectangle?	Each rectangle is 6 cm by 3 cm, so its perimeter is 18 cm.

Workbook Go further 5: Make up your own rectangle puzzle. Choose a rectangle with whole-cm sides, then give a partner only TWO clues about it (for example its area and its perimeter). Check that only ONE rectangle fits both clues. Which two clues did you choose, and why do they work?	Answers vary. Check that the clues fit the secret rectangle and that listing the rectangles for the first clue leaves exactly one for the second. Example: "Area 24 cm² and perimeter 20 cm". The rectangles with area 24 cm² are 24 by 1 (perimeter 50), 12 by 2 (28), 8 by 3 (22) and 6 by 4 (20), so only 6 by 4 fits.
Workbook Go further 6: Ali joins 5 same-size squares with sides of 2 cm, whole sides touching. What is the biggest perimeter his shape can have? The smallest?	Biggest 24 cm (12 sides of 2 cm, e.g. a row of 5). Smallest 20 cm (10 sides: a 2 by 2 square with one more square on a side).
Workbook Go further 7: A rectangle 7 cm by 5 cm has a 1 cm square cut out of each of its 4 corners. Find the new area and the new perimeter.	Area 35 − 4 = 31 cm². Perimeter still 24 cm: push the cut corners out and you get the 7 by 5 rectangle back.

These are the questions on the Think harder and Go further pages in the workbook. More are in the separate Question Bank.

72–80 min Talk time (Investigation debrief first)

DO Bring the group back together (camera on). First 3 minutes (72 to 75): the Investigation debrief. Ask for volunteers to share what they noticed, then show the model answer (in the Investigation row at the end of this lesson's answer key and in the notes of the Investigation slide). An unfinished Investigation can be shared at the start of next lesson. Then the talk question. Ask for volunteers first. A pass is always fine, and so is sharing with a partner. Do not call a child by name cold, and keep no public tallies or rankings. Nobody is called on by name. Come back to sneak-peek puzzle 2: "Now we know the idea, how could it help?"

SAY *"Two gardens have the same perimeter. Must they have the same area? Show me an example."*

"When does pushing out the steps NOT work?"

Discussion question	Listen for
Two gardens have the same perimeter. Must they have the same area? Give an example.	No. A 9 m by 1 m garden and a 5 m by 5 m garden both have a 20 m fence, but the areas are 9 m² and 25 m². Pushing out does not work when the shape has a dent: the two walls of the dent are extra.

80–90 min Exit ticket (80–86), look back (86–88) and closing (88–90)

DO Minutes 80 to 86. Give each child a slip of paper (or ask them to write on their journal page). No talking, no helping. Show the exit ticket slide. Children show the answer to you on camera in the open call (or keep it to share if they want); no photos and no private messages. Never mark or read answers out in front of the group.

Exit ticket	Answer

A rectangle has a perimeter of 12 cm. Its length is 4 cm. What is its area?	8 cm² (breadth 2 cm because 4 + 2 = 6, half of 12)

Look back (minutes 86 to 88, two minutes)

SAY *"Turn to the last page of today's lesson and find Look back. Tick or write one thing for each line: something you did well, the hardest part, and what helped you. There are no right answers, and you may pass on any line."*

"Would anyone like to share something they did well, or what helped? Thank you. When you got stuck and tried something else, that is exactly what mathematicians do."

DO Give a full 2 minutes. Never collect the page, mark it or read it out in front of the group: children keep it. You cannot see the paper. Tell the WHOLE group: "If you ticked still stuck, say check-in please in the open class, or ask your grown-up to tell me." Then talk it through with that child in the open session, with a parent nearby. No hand signals that single a child out, never a private message. Open the next lesson with the whole group: "What is still tricky from last time?" Invite volunteers only. Praise the strategy ("you drew it and that got you unstuck"), not the speed or the answer.

SAY *"You listed in order so nothing was missed, and you changed the shape to make it easy. Real architects do the same. Well done!"*

DO Minutes 88 to 90. Stamp the inquiry journal when a child has shown you their exit ticket. Mark S, D or H in the progress tracker at the back of this guide (it is for you only). If a child is marked D or H, plan two minutes of quiet catch-up at the start of the next lesson.

Extension for children who want more

Find all the rectangles with whole-cm sides and a perimeter of 24 cm. Which has the biggest area? (11 by 1, 10 by 2, 9 by 3, 8 by 4, 7 by 5, 6 by 6; the 6 by 6 square, 36 cm²)

Answer key (workbook)

Q1 A: 8 cm², 12 cm. B: 8 cm², 14 cm. C: 9 cm², 12 cm. D: 5 cm², 12 cm. (a) A and B (both 8 cm²). (b) A, C and D (all 12 cm).

Q2 Length + breadth = 9 m: 8 by 1 (8 m²), 7 by 2 (14 m²), 6 by 3 (18 m²), 5 by 4 (20 m²). 4 gardens. Biggest area: 5 m by 4 m, 20 m².

Q3 Rectangle perimeter 7 + 3 + 7 + 3 = 20 cm. Square side 20 ÷ 4 = 5 cm. Area 5 × 5 = 25 cm².

Q4 Missing sides 4 cm (10 − 6) and 3 cm (7 − 4). (a) Area 58 cm² (6 × 7 + 4 × 4 = 42 + 16, or 10 × 7 − 4 × 3 = 70 − 12). (b) Perimeter 34 cm (push the step out: 10 + 7 + 10 + 7).

Q5 (a) 10 cm² (b) 16 cm (c) The perimeter stays 16 cm (same as the 4 by 4 square); the area goes up by 6 cm² to 16 cm².

Q6 (a) Any square that touches the L along only one side, e.g. above the top square: perimeter 12 cm. (b) Only one place: the square to the right of the middle square (it touches 2 sides). (c) No. To make 8 cm the new square would have to touch 3 sides of the L, and there is no such gap.

Sneak-peek puzzles Puzzle 1: 6 squares; 10 cm round the edge Puzzle 2: Yes, e.g. 26 cm and 16 cm

Investigation (What if we change it? What stays the same?) Model answer and what to look for: 4 by 4: perimeter 16, area 16 becomes 5 by 3: perimeter 16, area 15. 6 cm: 24 and 36 become 7 by 5: 24 and 35. 10 cm: 40 and 100 become 11 by 9: 40 and 99. The perimeter stays the same and the area is always exactly 1 cm² smaller. This is true for every square (checked by computer for sides 2 to 499). Why: move the bottom row of 4 squares to the right side. The new column is only 3 squares tall, so 3 squares fit and 1 is left over. Look for: several tests, the pattern 'area drops by 1', and any picture or cut-and-move reason. Extension: what if we make it 2 cm longer and 2 cm shorter? (area drops by 4; perimeter still the same).

Story break try-it (Karen Uhlenbeck, optional) Rectangle: 6 x 4 = 24 square cm. Square: 5 x 5 = 25 square cm. Both have perimeter 20 cm, and the square has the larger area, by 1 square cm.

Go further (explore slide) 16 squares: make every rectangle. Which has the longest way round the edge? The shortest? Answer: 16 squares: 1 by 16 (34 cm round), 2 by 8 (20 cm), 4 by 4 (16 cm). Longest 1 by 16, shortest 4 by 4: the square, the most compact shape.

Lesson 8: Angle and Line Detectives

Week 4 (second lesson) • Angles, Perpendicular and Parallel Lines • Workbook: Lesson 8

Objective	Compare angles with a right angle using a folded-paper checker, count the angles at a point and the right angles in shapes and figures in an organised way, find a pattern in the counts, and find and draw perpendicular and parallel lines.
Thinking skills	Organised counting, pattern spotting, checking with a tool, explaining
Materials	Scrap paper to fold a right-angle checker, a ruler, mini whiteboards, square grid paper. For the sneak-peek puzzles: paper, a pencil and a few counters or small objects for each child.
Online set-up	Each child folds a right-angle checker on camera and goes on a 2-minute right-angle hunt at home; the tutor shares the grid slides to draw on. Print the workbook before the lesson. Pairs work in one breakout room for 10 minutes, then return for the check slide.
Prepare	Solve the worksheet yourself first. Set out one materials box per pair. Open the slides for this lesson and check the answers in the speaker notes.

0–2 min Opening

SAY *"Today you are angle and line detectives. Detectives do not guess: they check every clue with a tool. Our tool is a right-angle checker that you make yourself from any scrap of paper."*

"By the end you will be able to find every angle in a picture without missing one, and spot lines that never meet."

DO Greet each child by name. Show the lesson title slide. Read "Today we will…" aloud.

Ask the whole group once: "What is still tricky from last time? Did anyone try the Investigation at home? Volunteers can share one thing." Pass is fine.

2–10 min Sneak-peek puzzles (2–5), warm-up game (5–8) and quick questions (8–10)

SAY *"Here are two sneak-peek puzzles. These are ideas we have not learnt yet, so guessing is welcome and so is "I do not know". Nobody is marked. Write your answer on your own paper. When I say "show me", hold it up to the camera for me."*

DO Show the "Sneak-peek puzzles" slide (minutes 2 to 5). Read puzzle 1 aloud and give about 1 minute: it is an access puzzle, so any child can try it with objects, a drawing or counting. Then puzzle 2, about 1 minute, the thinking puzzle. Look at the papers on camera quietly. Do not show the answers on the slide and do not read answers out. Say "thank you" to everyone in the same way.

Nobody is labelled. Never say who got it, who is on which page, or "fast group" or "top group". Do not single anyone out. Invite the whole group once Start here is done. It is an invitation, never a ranking: every child can choose Think harder or the Investigation once page 1 and Q4 and Q5 of page 2 are done. We come back to puzzle 2 in talk time, for volunteers.

Sneak-peek puzzle	Answer (tutor only) and what to listen for
Puzzle 1 (access): Look at the letters L, H and O. Which letters have a square corner? Which letter has two lines that never meet, however far they go?	L and H have square corners; H has two lines that never meet. Listen for: L and H have square corners; O has none. The two upright lines of H stay the same distance apart.
Puzzle 2 (thinking): Draw a rectangle. Draw a straight line from one corner to the opposite corner. You get two triangles. How many right angles does each triangle have? How do you know?	1 right angle each. Listen for: Only the corner that was not cut is still a right angle. The corners that were cut are split into two smaller angles.

Choice during the lesson (groups of 8, online)

- Minutes 10 to 40 (explore, mini-lesson, brain break): everyone does the same. The explore slide has a Go further what-if from minute 10, so a pair that has finished its first task has real work at once. Do not say who is ready for it.
- Minute 40: everyone opens Start here page 1 and works in pairs. A line at the top says: “Do page 1, then Q4 and Q5 on page 2. After that you may choose: Think harder, Go further or the Investigation.” You never announce who is on which page. Invite the whole group, never one child: “When you have finished Start here you can choose Think harder, Go further or the Investigation.” Do not split a pair unless both children agree.
- The first-step help for the Investigation is printed on page 2 of Start here, so every child sees it. Pairs work in breakout rooms, so visit EVERY room by minute 47 (about a minute each, camera on, a parent nearby) and ask “how do you know?”, not “is it done?”. At about minute 52 check in again with two or three children, including children on Start here.
- Minute 67: everyone chooses what to do next: finish page 2, Think harder, Go further or the Investigation. The Investigation has its own slot, minutes 67 to 72, and its debrief opens talk time. Everyone is back together for talk time at minute 72. An Investigation that is not finished goes home; volunteers share one thing at the start of next lesson.
- Next lesson everything starts again from the same pages: nobody is on a track.

DO Warm-up game (minutes 5 to 8). Now the game, then the quick questions (minutes 8 to 10): Clock angles. Make the clock hands with your arms: is the angle at 3, 1, 5, 9, 10 and 4 o’clock a right angle, smaller or bigger?

SAY *“Look at each clock. Make the two hands with your arms. Is the angle a right angle, like the corner of a book? Smaller? Or bigger?”*

“Clock 1: 3 o’clock. Show me!”

DO After the game, ask the quick questions on the warm-up slides. Use thumbs or a whisper to partners before taking answers.

Ask	Expect
How many right angles does a rectangle have?	4
How many sides does a hexagon have?	6

Name a capital letter made of 2 lines that meet at a right angle.	L (or T)
Do the two long edges of a ruler ever meet?	No: they are parallel

10–25 min Explore (in pairs, with objects)

SAY *"I will fold a torn piece of paper with wiggly edges, twice. Will the corner I make be a right angle, even though the paper is wiggly? Thumbs up for yes, down for no."*

DO 1. Take any scrap of paper.

2. Fold it in half to make a straight fold.

3. Fold again so the fold lies exactly on itself.

4. Use the corner to hunt for right angles, smaller angles and bigger angles.

SAY *"Talk to your partner while you work. I will come round and ask you questions."*

IF STUCK Stuck folding: fold once to make a straight edge, then fold that edge onto itself so the two halves line up exactly.

Not sure if it is a right angle: put the corner of your checker into the angle. If both arms line up with the edges, it is a right angle.

Ready for more: find an angle that is exactly 2 right angles. (A straight line!)

DO Pairs work from minute 10. If a pair finishes the first task, they may take the Go further what-if on the slide (do not say who). At about minute 15 bring the group back together and take the three questions below, one at a time (minutes 15 to 25).

Ask	Expect
Why is the folded corner always a right angle?	The first fold makes a straight line. Folding it onto itself cuts the straight line into 2 equal angles: each is a right angle.
Is an angle bigger if its arms are longer?	No. The size is how far the arms open, not how long they are.
Where did you find angles bigger than a right angle?	For example an open laptop, a door opened wide, a clock at 5 o'clock.

25–35 min Mini-lesson (you teach, they join in)

DO Show the mini-lesson slides. Think aloud as you work. Say what you notice, what you test and how you check. Children copy how you think.

SAY *"Here are 4 lines from one point. I count the small angles first: 3. Then the angles made of 2 small ones: 2. Then 3 small ones: 1. 3 + 2 + 1 = 6. I go in order so I never miss one."*

"2 lines make 1 angle, 3 lines make 3, 4 lines make 6. The jumps are 2, 3, 4… so the next jump is 5. I can predict without drawing."

"To count right angles in a picture I visit every point where lines meet, one by one, and write how many right angles are at each point. Then I add."

Worked example on the slides	Answer
Count the angles at a point. 4 lines from one point make 3 small angles. Count in order: made of 1 small angle: 3. Made of 2: 2. Made of 3: 1. Look for the pattern: 1, 3, 6. What comes next?	4 lines: 3 + 2 + 1 = 6 angles. The jumps are 2, 3, 4, so 5 lines make 10.
Right angles where lines meet. A rectangle has one line from the middle of the top to the middle of the bottom. Visit every point where lines meet: 4 corners have 1 right angle each; the 2 middle points have 2 each. How many right angles altogether?	4 + 4 = 8 right angles

SAY *"The key idea today is: count in order, look for a pattern. Visit every corner or meeting point, and count small angles first. When the counts grow, look for a pattern. Say it with me."*

DO Show the key idea slide at minutes 34 to 35. The "common mistakes" on that slide are for you, not the children. Watch for them during the worksheet:

- Thinking an angle with longer arms is bigger, or that a turned right angle is not a right angle.
- Counting every corner of a shape as a right angle, even a dent corner (inside it is bigger than a right angle).
- Counting angles or line pairs at random, so some are missed or counted twice.

35–40 min Brain break

DO Body angles. Make a right angle with your arms, then a smaller angle, then a bigger one. Now hold your arms parallel, then perpendicular.

Water. Toilet break for anyone who needs it. Keep it to five minutes.

DO Story break (optional, 2 minutes): after the movement, show the Story break slide (Ibn al-Haytham, Born probably in Basra (now Iraq); worked mostly in Cairo, Egypt, about 965 to about 1040) and read the story aloud. Link to today: Today you look for angles and perpendicular lines, and light bouncing off a mirror makes equal angles on either side of the perpendicular. Then offer the try-it as a question for everyone to think about: "undefined" Anyone may pass, nobody is called on by name and nothing is marked. The answer is in the notes of the slide and in the answer key below. Skip it if you are short of time: click straight on to the worksheet.

DO Accuracy note for you (do not read out): The house-arrest story is a legend (MacTutor says it 'appears' true; Wikipedia calls it a legend). Birth place and death place are uncertain, and sources give 1039 or 1040. The equal-angles rule is the modern law of reflection and is not claimed as his own words.

40–72 min Worksheet (Start here pages, then your choice, with its own Investigation slot at 67–72)

DO Minute 40: give out the workbook. Read the title of Lesson 8 aloud. Say "Pairs first. Talk before you write."

Minutes 40 to 52: page 1 (Start here) in pairs; where the deck has a short teaching slide, it comes first. Pairs are in breakout rooms, so visit EVERY room by minute 47 (about a minute each; camera on, parent nearby), then check in again at about minute 52. Minutes 52 to 54: check page 1 together. Minutes 54 to 57: short teaching before page 2. Minutes 57 to 65: each child works alone on page 2. Minutes 65 to 67: check page 2 together. Minutes 67 to 72: everyone chooses Think harder, Go further or the Investigation (the Investigation has its own slide, minutes 67 to 72). A child who finishes early starts on one of them at once, and any child may choose them once page 1 and Q4 and Q5 of page 2 are done.

SAY *"Pairs first. Read each question aloud and talk about it before you write. I am not looking for speed. I am looking for good thinking."*

DO Circulate and ask:

- Ask: "Did you check that corner with your checker, or did you guess?"
- Ask: "Which points have you visited? Which are left?"
- Ask: "Do these two lines ever meet if we make them longer?"

IF STUCK Hint 1: "Use your right-angle checker on every corner."

Hint 2: "Count the small angles first, then the bigger ones made of small ones."

Hint 3: "Make a list of every pair of lines and test each pair."

DO Anyone who wants more may choose Think harder first, then Go further, or the Investigation (its easy first step is on page 2 of Start here). Invite the whole group: "When you have finished Start here you can choose Think harder, Go further or the Investigation." Never announce who is on which page, and do not split a pair unless both children agree.

Think harder and Go further questions

Question	Answer
Workbook Think harder 1: I have 4 sides and 4 right angles. My sides are not all the same length. What shape am I? How many pairs of parallel lines do I have?	A rectangle. 2 pairs of parallel lines.
Workbook Think harder 2: A U shape is made of 5 squares: 3 squares in a row with 1 more square on top of each end square. How many right angles are inside it?	6 (it has 8 corners; the 2 corners at the bottom of the gap are bigger than a right angle inside the shape).
Workbook Think harder 3: Look at a clock face. For each o'clock time from 1 o'clock to 11 o'clock, is the angle between the hands (the short way round) smaller than, the same as, or bigger than a right angle? Put each time in the right group.	Smaller than a right angle: 1, 2, 10 and 11 o'clock. A right angle: 3 and 9 o'clock. Bigger: 4, 5, 6, 7 and 8 o'clock (at 6 o'clock the hands make a straight line).

Workbook Think harder 4: The # sign is made of 4 lines: 2 upright lines and 2 lines across. How many pairs of parallel lines does it have? How many pairs of perpendicular lines?	2 pairs of parallel lines and 4 pairs of perpendicular lines.
Workbook Go further 5: Make up your own "Which shape am I?" riddle about a shape on square grid paper. Use words like sides, right angles, parallel and perpendicular, and give 3 clues. Draw the answer. Then check: can a partner draw a DIFFERENT shape that fits your clues? If yes, add a clue.	Answers vary. Check that the drawn shape fits every clue and that the clues are about sides, right angles and parallel lines. Example: "I have 6 straight sides. I have 5 right angles inside me. I have one dent." (An L shape.) Other shapes could fit the first two clues, so the third clue matters.
Workbook Go further 6: A long rectangle is cut into 6 squares in a row by 5 lines. How many right angles can you find?	24. 4 at the corners, and each of the 5 lines makes 2 right angles at the top and 2 at the bottom: 4 + 5 × 4 = 24.
Workbook Go further 7: Can a 6-sided shape have 6 right angles inside it? What is the most right angles a 6-sided shape can have inside?	No. The most is 5, for example an L shape. Children try to draw 6 square corners and always find a dent. A proof is not expected.

These are the questions on the Think harder and Go further pages in the workbook. More are in the separate Question Bank.

72–80 min Talk time (Investigation debrief first)

DO Bring the group back together (camera on). First 3 minutes (72 to 75): the Investigation debrief. Ask for volunteers to share what they noticed, then show the model answer (in the Investigation row at the end of this lesson's answer key and in the notes of the Investigation slide). An unfinished Investigation can be shared at the start of next lesson. Then the talk question. Ask for volunteers first. A pass is always fine, and so is sharing with a partner. Do not call a child by name cold, and keep no public tallies or rankings. Nobody is called on by name. Come back to sneak-peek puzzle 2: "Now we know the idea, how could it help?"

SAY *"Mei says an angle with longer arms is bigger. Do you agree? How can you test it?"*

"How did you make sure you did not miss any angles?"

Discussion question	Listen for
Mei says: "An angle with longer arms is a bigger angle." Do you agree? How can you test it?	No. The size of an angle is how far the arms open (how much one arm turns to reach the other). Draw two angles that open the same amount, one with long arms and one with short arms: the checker fits both the same way. To miss none, count in order: small angles first, or visit every meeting point.

80–90 min Exit ticket (80–86), look back (86–88) and closing (88–90)

DO Minutes 80 to 86. Give each child a slip of paper (or ask them to write on their journal page). No talking, no helping. Show the exit ticket slide. Children show the answer to you on camera in the

open call (or keep it to share if they want); no photos and no private messages. Never mark or read answers out in front of the group.

Exit ticket	Answer
Which of these letters have BOTH a pair of parallel lines and a pair of perpendicular lines: F, V, H, Z?	F and H

Look back (minutes 86 to 88, two minutes)

SAY *"Turn to the last page of today's lesson and find Look back. Tick or write one thing for each line: something you did well, the hardest part, and what helped you. There are no right answers, and you may pass on any line."*

"Would anyone like to share something they did well, or what helped? Thank you. When you got stuck and tried something else, that is exactly what mathematicians do."

DO Give a full 2 minutes. Never collect the page, mark it or read it out in front of the group: children keep it. You cannot see the paper. Tell the WHOLE group: "If you ticked still stuck, say check-in please in the open class, or ask your grown-up to tell me." Then talk it through with that child in the open session, with a parent nearby. No hand signals that single a child out, never a private message. Open the next lesson with the whole group: "What is still tricky from last time?" Invite volunteers only. Praise the strategy ("you drew it and that got you unstuck"), not the speed or the answer.

SAY *"You checked with a tool, counted in order and found a pattern. That is how real detectives work. Well done!"*

DO Minutes 88 to 90. Stamp the inquiry journal when a child has shown you their exit ticket. Mark S, D or H in the progress tracker at the back of this guide (it is for you only). If a child is marked D or H, plan two minutes of quiet catch-up at the start of the next lesson.

Extension for children who want more

How many right angles are in a square drawn as a noughts-and-crosses grid (2 lines down and 2 lines across inside it)? (36)

Answer key (workbook)

Q1 Smaller than a right angle: b, f. Right angle: a, d. Bigger than a right angle: c, e.

Q2 A 8 (given as the example; the 4 dent corners are bigger), B 6 (2 dent corners), C 3 (the 2 bottom corners and the top of the roof), D 2.

Q3 (a) 5 + 4 + 3 + 2 + 1 = 15 angles. (b) 8 lines: 7 + 6 + 5 + 4 + 3 + 2 + 1 = 28 angles.

Q4 (a) E and H: both. T and L: perpendicular only. Z and N: parallel only. (b) E: 3 pairs of parallel lines and 3 pairs of perpendicular lines.

Q5 (a) A line through C going 1 square right and 1 square up at each step (the same steps as AB). (b) A line through D going 1 square right and 1 square down at each step. (c) A line through G going 1 square right and 2 squares down (or 1 left and 2 up) at each step. Check each with the right-angle checker.

Q6 16 right angles: 4 outer corners × 1, 4 points where the lines meet the sides × 2, and 4 in the middle.

Sneak-peek puzzles Puzzle 1: L and H have square corners; H has two lines that never meet Puzzle 2: 1 right angle each

Investigation (Staircase corners: find the pattern) Model answer and what to look for: Steps 1 to 4: right angles 4, 5, 6, 7; bigger than a right angle 0, 1, 2, 3; all corners 4, 6, 8, 10. So right-angle corners = steps + 3, bigger corners = steps − 1, all corners = 2 × steps + 2. 10 steps: 13, 9 and 22 (checked by computer by building each staircase). The total is always even because each new step adds 2 corners (one right angle and one dent). Look for: careful organised counting, a table, a stated rule, and a prediction that can be checked. Any pair can draw the 10-step staircase to test the prediction.

Story break try-it (Ibn al-Haytham, optional) 90 - 30 = 60 degrees with the perpendicular. The reflected ray also makes 60 degrees, so the angle between the two rays is 60 + 60 = 120 degrees.

Go further (explore slide) Fold your corner in half again. Smaller or bigger than a right angle? Find two. Answer: Smaller than a right angle: it is half of a right angle, so two of them make one right angle. It appears at the corner of a set square, or when a square is cut along its diagonal.

Lesson 9: Measure and Money Masters

Week 5 (first lesson) • Length, mass, volume, money and time • Workbook: Lesson 9

Objective	Change compound units (m and cm, kg and g, ℓ and ml, $ and ¢) to one unit to compare, add and subtract; plan a budget; find durations and start and end times by drawing a time line, in 12-hour and 24-hour time.
Thinking skills	Converting units, comparing, budgeting, drawing a time line, working backwards, explaining
Materials	Party menu cards (one per pair), play money, a measuring tape or metre ruler, a kitchen scale and a measuring jug if available, a clock with movable hands, mini whiteboards. For the sneak-peek puzzles: paper, a pencil and a few counters or small objects for each child.
Online set-up	Show the menu card on screen; children use paper and pencil (play money optional). Pairs work in breakout rooms and type their best party plan in the chat.
Prepare	Solve the worksheet yourself first. Set out one materials box per pair. Open the slides for this lesson and check the answers in the speaker notes.

0–2 min Opening

SAY *"A tailor, a baker, a pharmacist and a bus captain all measure things every day: metres and centimetres, kilograms and grams, litres and millilitres, dollars and cents, hours and minutes. Today we become Measure and Money Masters. Our two tricks: change to one unit, and draw a time line."*

DO Greet each child by name. Show the lesson title slide. Read "Today we will…" aloud.

Ask the whole group once: "What is still tricky from last time? Did anyone try the Investigation at home? Volunteers can share one thing." Pass is fine.

2–10 min Sneak-peek puzzles (2–5), warm-up game (5–8) and quick questions (8–10)

SAY *"Here are two sneak-peek puzzles. These are ideas we have not learnt yet, so guessing is welcome and so is "I do not know". Nobody is marked. Write your answer on your own paper. When I say "show me", hold it up to the camera for me."*

DO Show the "Sneak-peek puzzles" slide (minutes 2 to 5). Read puzzle 1 aloud and give about 1 minute: it is an access puzzle, so any child can try it with objects, a drawing or counting. Then puzzle 2, about 1 minute, the thinking puzzle. Look at the papers on camera quietly. Do not show the answers on the slide and do not read answers out. Say "thank you" to everyone in the same way.

Nobody is labelled. Never say who got it, who is on which page, or "fast group" or "top group". Do not single anyone out. Invite the whole group once Start here is done. It is an invitation, never a ranking: every child can choose Think harder or the Investigation once page 1 and Q4 and Q5 of page 2 are done. We come back to puzzle 2 in talk time, for volunteers.

Sneak-peek puzzle	Answer (tutor only) and what to listen for

Puzzle 1 (access): Which is longer: 1 m or 90 cm? Say how you know.	1 m. Listen for: 1 m is 100 cm, and 100 is more than 90.
Puzzle 2 (thinking): A film starts at 3:45 p.m. and lasts 1 hour 30 minutes. When does it end? Say how you counted.	5:15 p.m. Listen for: 3:45 + 1 h = 4:45. 15 minutes more makes 5:00, and 15 more makes 5:15.

Choice during the lesson (groups of 8, online)

- Minutes 10 to 40 (explore, mini-lesson, brain break): everyone does the same. The explore slide has a Go further what-if from minute 10, so a pair that has finished its first task has real work at once. Do not say who is ready for it.
- Minute 40: everyone opens Start here page 1 and works in pairs. A line at the top says: “Do page 1, then Q4 and Q5 on page 2. After that you may choose: Think harder, Go further or the Investigation.” You never announce who is on which page. Invite the whole group, never one child: “When you have finished Start here you can choose Think harder, Go further or the Investigation.” Do not split a pair unless both children agree.
- The first-step help for the Investigation is printed on page 2 of Start here, so every child sees it. Pairs work in breakout rooms, so visit EVERY room by minute 47 (about a minute each, camera on, a parent nearby) and ask “how do you know?”, not “is it done?”. At about minute 52 check in again with two or three children, including children on Start here.
- Minute 67: everyone chooses what to do next: finish page 2, Think harder, Go further or the Investigation. The Investigation has its own slot, minutes 67 to 72, and its debrief opens talk time. Everyone is back together for talk time at minute 72. An Investigation that is not finished goes home; volunteers share one thing at the start of next lesson.
- Next lesson everything starts again from the same pages: nobody is on a track.

DO Warm-up game (minutes 5 to 8). Now the game, then the quick questions (minutes 8 to 10): Name that unit. The tutor names an object and a number; write the best unit: m, cm, km, g, kg, ml or ℓ.

SAY *“I will name an object and a number. You write the best unit on your whiteboard: m, cm, km, g, kg, ml or ℓ. Ready? A bus is 12 what long? Hold up! Now tell your partner why it is not 12 cm or 12 km.”*

DO After the game, ask the quick questions on the warm-up slides. Use thumbs or a whisper to partners before taking answers.

Ask	Expect
1 m 40 cm = __ cm	140 cm
3 kg = __ g	3000 g
2 ℓ 50 ml = __ ml	2050 ml
1 hour 20 minutes = __ minutes	80 minutes

2:00 p.m. in 24-hour time is __	14:00

10–25 min Explore (in pairs, with objects)

SAY *"Look at the menu card. Before you add anything: do you think you can buy pizza, wings and cupcakes with $20? Show thumbs up or down, then we will check."*

DO 1. You have $20 to plan a party for 8 friends.

2. Each friend drinks 250 ml of punch. Buy at least 2 ℓ of punch (any number of bottles).

3. Buy at least one food. You may buy more than one of an item. Spend as close to $20 as you can.

4. Go further: can you spend exactly $20?

SAY *"Talk to your partner while you work. I will come round and ask you questions."*

IF STUCK Stuck on the punch: "How many millilitres do 8 friends drink? How many ml is 2 ℓ?"

Adding prices: "Change each price to cents, or add the dollars first and then the cents."

Over $20: "Which item could you swap for a cheaper one?"

Ready for more: "Can you spend EXACTLY $20? Try two pizzas, or three punches and two cupcakes."

DO Pairs work from minute 10. If a pair finishes the first task, they may take the Go further what-if on the slide (do not say who). At about minute 15 bring the group back together and take the three questions below, one at a time (minutes 15 to 25).

Ask	Expect
How much punch do 8 friends need? How many bottles?	8 × 250 ml = 2000 ml = 2 ℓ, so at least 1 bottle. You may buy more.
Who got closest to $20? How did you add the prices?	Open. Changing to cents, or adding dollars then cents, both work.
Can you spend exactly $20?	Yes, 15 ways with up to 4 of each item, for example pizza + 2 punch + cupcake; 2 pizzas + punch; 3 punch + 2 cupcakes; wings + 2 curry puffs + 2 punch.

25–35 min Mini-lesson (you teach, they join in)

DO Show the mini-lesson slides. Think aloud as you work. Say what you notice, what you test and how you check. Children copy how you think.

SAY *"2 m 35 cm. 1 m is 100 cm, so 2 m is 200 cm. 200 and 35 more is 235 cm. Careful with 2 m 5 cm: that is 200 and 5, so 205 cm, not 25."*

"The flour bag has 2 kg 100 g. I change it to grams: 2100 g. Mum uses 850 g. 2100 – 850 = 1250 g. Now I change back: 1000 g is 1 kg, so 1 kg 250 g. I check: 1250 + 850 = 2100."

"From 9:45 to 11:20. Time does not work like ordinary numbers, because an hour is 60 minutes, not 100. So I draw a time line and jump to the next o'clock: 15 minutes to 10:00, 1 hour to 11:00,

20 minutes to 11:20. Altogether 1 hour 35 minutes."

Worked example on the slides	Answer
Change to one unit. A bag has 2 kg 100 g of flour. Mum uses 850 g. How much flour is left?	2 kg 100 g = 2100 g. 2100 – 850 = 1250 g = 1 kg 250 g.
Draw a time line. A movie starts at 2:50 p.m. and lasts 1 h 45 min. When does it end?	2:50 + 10 min = 3:00, + 1 h = 4:00, + 35 min = 4:35 p.m.

SAY *"The key idea today is: one unit, and a time line. Change to one unit before you compare, add or subtract. For time, draw a time line and jump to the next o'clock. Say it with me."*

DO Show the key idea slide at minutes 34 to 35. The "common mistakes" on that slide are for you, not the children. Watch for them during the worksheet:

- Writing the small unit straight after the big one: 2 m 5 cm as 25 cm, 3 kg 40 g as 340 g (it is 205 cm and 3040 g).
- Treating time like ordinary numbers: 9:45 to 11:20 worked out as 11.20 – 9.45 = 1.75, or counting 100 minutes in an hour.
- In the 24-hour clock, adding 12 to a.m. times or to 12:30 p.m., or reading 17:00 as 7 p.m.

35–40 min Brain break

DO Human clock. Stretch one arm as the hour hand and one as the minute hand. Show 3 o'clock, 6 o'clock, quarter past 9 … and 15:00!

Water. Toilet break for anyone who needs it. Keep it to five minutes.

DO Story break (optional, 2 minutes): after the movement, show the Story break slide (Zu Chongzhi, China (born in Jiankang, now Nanjing), 429 to about 500 (sources differ)) and read the story aloud. Link to today: Today you work with units of time, and Zu measured the year in days, hours and parts of a day. Then offer the try-it as a question for everyone to think about: "undefined" Anyone may pass, nobody is called on by name and nothing is marked. The answer is in the notes of the slide and in the answer key below. Skip it if you are short of time: click straight on to the worksheet.

DO Accuracy note for you (do not read out): Wikipedia and MacTutor disagree on his dates and on the polygon he used, so the story gives neither polygon nor an exact death year. The 'within about a minute' comparison is our own check: 365.24281481 - 365.24219878 days is about 53 seconds, and MacTutor says about 50 seconds. Only MacTutor says the calendar was never put into use, so the story leaves it out.

40–72 min Worksheet (Start here pages, then your choice, with its own Investigation slot at 67–72)

DO Minute 40: give out the workbook. Read the title of Lesson 9 aloud. Say "Pairs first. Talk before you write."

Minutes 40 to 52: page 1 (Start here) in pairs; where the deck has a short teaching slide, it comes first. Pairs are in breakout rooms, so visit EVERY room by minute 47 (about a minute each; camera on, parent nearby), then check in again at about minute 52. Minutes 52 to 54: check page 1

together. Minutes 54 to 57: short teaching before page 2. Minutes 57 to 65: each child works alone on page 2. Minutes 65 to 67: check page 2 together. Minutes 67 to 72: everyone chooses Think harder, Go further or the Investigation (the Investigation has its own slide, minutes 67 to 72). A child who finishes early starts on one of them at once, and any child may choose them once page 1 and Q4 and Q5 of page 2 are done.

SAY *"Pairs first. Read each question aloud and talk about it before you write. I am not looking for speed. I am looking for good thinking."*

DO Circulate and ask:

- Ask: "What did you change the units to? Why that unit?"
- Ask: "Show me your time line. Where did you jump to the o'clock?"
- Ask: "How do you know your change is right? Add it back on."

IF STUCK Hint 1: "Change everything to the smaller unit first: cm, g, ml, ¢ or minutes."

Hint 2: "Draw a time line. Jump to the next o'clock, then whole hours, then the minutes left."

Hint 3: "For the 24-hour clock: after 12 noon, add 12 to the hour."

DO Anyone who wants more may choose Think harder first, then Go further, or the Investigation (its easy first step is on page 2 of Start here). Invite the whole group: "When you have finished Start here you can choose Think harder, Go further or the Investigation." Never announce who is on which page, and do not split a pair unless both children agree.

Think harder and Go further questions

Question	Answer
Workbook Think harder 1: A 2 ℓ bottle of juice is poured into cups. Each cup holds 300 ml. How many cups can be filled? How much juice is left over?	6 cups, 200 ml left (300, 600, 900, 1200, 1500, 1800 ml; 2000 − 1800 = 200 ml)
Workbook Think harder 2: A shop opens at 09:30 and closes at 21:00. How long is it open?	11 h 30 min (30 min to 10:00, then 11 h to 21:00)
Workbook Think harder 3: A box with 4 books weighs 1 kg 50 g. The empty box weighs 250 g. All the books weigh the same. How heavy is one book?	200 g (1050 g − 250 g = 800 g; 800 ÷ 4 = 200 g)
Workbook Think harder 4: Buses leave the interchange every 25 minutes. The first bus leaves at 06:40. The ride to school takes 35 minutes. Ravi must reach school by 08:30. Which is the latest bus he can take?	The 07:55 bus (buses: 06:40, 07:05, 07:30, 07:55, 08:20; 07:55 + 35 min = 08:30; the 08:20 bus arrives at 08:55, too late).
Workbook Go further 5: A bottle full of water weighs 1 kg 200 g. After half of the water is poured out, it weighs 750 g. How heavy is the empty bottle?	300 g (half the water = 1200 − 750 = 450 g; all the water = 900 g; bottle = 1200 − 900 = 300 g)

Workbook Go further 6: Mei buys 3 pens and 2 files for $9.20. Each file costs 60¢ more than each pen. How much does one pen cost?	$1.60 (take away the extra 2 × 60¢ = $1.20: $8.00 for 5 pens; 800¢ ÷ 5 = 160¢. A file is $2.20.)
Workbook Go further 7: Make up your own problem about money or time that needs at least 2 steps. Write it, solve it and check it. Then change ONE number and solve it again. What changes in the answer, and why? Swap with a partner.	Answers vary. Check that the problem needs 2 or more steps, the answer is right, and the child can say what changed. Example: "Mei has $20. She buys a $7.85 book and a $5.50 water bottle. Can she also buy 3 stickers at $2.35 each?" She has $20 − $13.35 = $6.65 left, but 3 stickers cost $7.05, so she is 40¢ short. If she had $25 she would have $11.65 left, enough for the stickers, with $4.60 change.

These are the questions on the Think harder and Go further pages in the workbook. More are in the separate Question Bank.

72–80 min Talk time (Investigation debrief first)

DO Bring the group back together (camera on). First 3 minutes (72 to 75): the Investigation debrief. Ask for volunteers to share what they noticed, then show the model answer (in the Investigation row at the end of this lesson's answer key and in the notes of the Investigation slide). An unfinished Investigation can be shared at the start of next lesson. Then the talk question. Ask for volunteers first. A pass is always fine, and so is sharing with a partner. Do not call a child by name cold, and keep no public tallies or rankings. Nobody is called on by name. Come back to sneak-peek puzzle 2: "Now we know the idea, how could it help?"

SAY *"Which is longer: 3 m 5 cm or 350 cm? How do you know?"*

"Why can we not just do 11.20 − 9.45 to find how long it is from 9:45 to 11:20?"

Discussion question	Listen for
Which is longer: 3 m 5 cm or 350 cm? How do you know? Why is 2 m 5 cm not 25 cm?	350 cm is longer: 3 m 5 cm = 305 cm. Time is not like ordinary numbers: an hour has 60 minutes, not 100, so we count on with a time line (15 min + 1 h + 20 min = 1 h 35 min).

80–90 min Exit ticket (80–86), look back (86–88) and closing (88–90)

DO Minutes 80 to 86. Give each child a slip of paper (or ask them to write on their journal page). No talking, no helping. Show the exit ticket slide. Children show the answer to you on camera in the open call (or keep it to share if they want); no photos and no private messages. Never mark or read answers out in front of the group.

Exit ticket	Answer
A film starts at 3:30 p.m. and lasts 1 h 20 min. When does it end?	4:50 p.m. (3:30 + 30 min = 4:00, + 50 min = 4:50)

Look back (minutes 86 to 88, two minutes)

SAY *"Turn to the last page of today's lesson and find Look back. Tick or write one thing for each line: something you did well, the hardest part, and what helped you. There are no right answers, and you may pass on any line."*

"Would anyone like to share something they did well, or what helped? Thank you. When you got stuck and tried something else, that is exactly what mathematicians do."

DO Give a full 2 minutes. Never collect the page, mark it or read it out in front of the group: children keep it. You cannot see the paper. Tell the WHOLE group: "If you ticked still stuck, say check-in please in the open class, or ask your grown-up to tell me." Then talk it through with that child in the open session, with a parent nearby. No hand signals that single a child out, never a private message. Open the next lesson with the whole group: "What is still tricky from last time?" Invite volunteers only. Praise the strategy ("you drew it and that got you unstuck"), not the speed or the answer.

SAY *"Today you changed to one unit to compare and calculate, you planned a budget, and you drew time lines to count across the hour. Measure and Money Masters!"*

DO Minutes 88 to 90. Stamp the inquiry journal when a child has shown you their exit ticket. Mark S, D or H in the progress tracker at the back of this guide (it is for you only). If a child is marked D or H, plan two minutes of quiet catch-up at the start of the next lesson.

Extension for children who want more

Plan a real outing (a trip to the zoo or a hawker lunch): find the times on a timetable, the total cost for your family and the change from a $50 note.

Answer key (workbook)

Q1 (a) 200 cm and 3000 g (b) 306 cm and 2450 g (c) 4 ℓ 80 ml (d) 4 × 65 cm = 260 cm; 300 cm − 260 cm = 40 cm

Q2 (a) 1 kg, 1 kg 3 g, 1030 g, 1 kg 300 g (b) 1 kg is 1000 g, so 1 kg 30 g = 1030 g, not 130 g.

Q3 No: $7.85 + $6.90 + $5.50 = $20.25, which is 25¢ more than $20.

Q4 (a) 07:15, 15:40, 8:05 p.m., 12:30 (b) 17 − 12 = 5, so 17:00 is 5 p.m., not 7 p.m.

Q5 (a) 1 h 25 min (15 min + 1 h + 10 min) (b) 5:25 p.m. (4:35 + 25 min = 5:00, + 25 min = 5:25) (c) 7:25 p.m. (9:15 back 1 h = 8:15, back 50 min = 7:25)

Sneak-peek puzzles Puzzle 1: 1 m Puzzle 2: 5:15 p.m.

Investigation (Find all the ways to pay) Model answer and what to look for: There are 12 ways (20c, 10c, 5c coins): 2,1,0; 2,0,2; 1,3,0; 1,2,2; 1,1,4; 1,0,6; 0,5,0; 0,4,2; 0,3,4; 0,2,6; 0,1,8; 0,0,10. Fewest coins: two 20c and one 10c (3 coins). Most: ten 5c coins (10 coins). Systematic method: fix the number of 20c coins (2, then 1, then 0), then the number of 10c coins, then fill with 5c coins. Look for: an organised list, no repeats (the same set of coins in a different order is one way), and the reason for being sure. Extension: what if we also have a 50c coin? (13 ways).

Story break try-it (Zu Chongzhi, optional) A quarter of a day is 24 / 4 = 6 hours. In 4 years: 4 x 365 = 1460 days, plus 4 quarters = 1 day, so 1461 days.

Go further (explore slide) 12 friends and $30: how many bottles of punch? Can you spend exactly $30? Answer: 12 x 250 ml = 3000 ml = 3 l, so 2 bottles (2 l each) at least. Yes, there are many ways, e.g. 2 punch + 3 cupcakes + 3 balloons ($6 + $16.50 + $7.50), or 2 curry puffs + 2 punch + 2 cupcakes + 2 balloons.

Lesson 10: Challenge Day

Week 5 (second lesson) • Mixed puzzles: bar graphs and logic • Workbook: Lesson 10

Objective	Choose a way to start a puzzle (draw it, list it, make a table), solve a logic grid with ticks and crosses, read and complete a bar graph with a scale, and explain the thinking.
Thinking skills	Choosing a strategy, logical reasoning, reading scales, listing systematically, perseverance, explaining
Materials	Puzzle cards for 2 stations (list, measure), digit cards 0 to 9 (or paper cards), squared paper, mini whiteboards, certificates, stamps. For the sneak-peek puzzles: paper, a pencil and a few counters or small objects for each child.
Online set-up	Two stations, 7 minutes each. Pairs stay in one breakout room, and the tutor shares the next station card when the timer ends. No physical materials beyond paper. Logic grids and graphs are shared on screen and annotated together.
Prepare	Solve the worksheet yourself first. Set out one materials box per pair. Open the slides for this lesson and check the answers in the speaker notes.

0–2 min Opening

SAY *"Welcome to Challenge Day! Today there is no new topic: there are puzzles from every lesson. Clever inquirers do not always know the answer straight away. They know how to START: draw it, list it, or make a table."*

DO Greet each child by name. Show the lesson title slide. Read "Today we will…" aloud.

Ask the whole group once: "What is still tricky from last time? Did anyone try the Investigation at home? Volunteers can share one thing." Pass is fine.

2–10 min Sneak-peek puzzles (2–5), warm-up game (5–8) and quick questions (8–10)

SAY *"Here are two sneak-peek puzzles. These are ideas we have not learnt yet, so guessing is welcome and so is "I do not know". Nobody is marked. Write your answer on your own paper. When I say "show me", hold it up to the camera for me."*

DO Show the "Sneak-peek puzzles" slide (minutes 2 to 5). Read puzzle 1 aloud and give about 1 minute: it is an access puzzle, so any child can try it with objects, a drawing or counting. Then puzzle 2, about 1 minute, the thinking puzzle. Look at the papers on camera quietly. Do not show the answers on the slide and do not read answers out. Say "thank you" to everyone in the same way.

Nobody is labelled. Never say who got it, who is on which page, or "fast group" or "top group". Do not single anyone out. Invite the whole group once Start here is done. It is an invitation, never a ranking: every child can choose Think harder or the Investigation once page 1 and Q4 and Q5 of page 2 are done. We come back to puzzle 2 in talk time, for volunteers.

Sneak-peek puzzle	Answer (tutor only) and what to listen for

Puzzle 1 (access): Ali, Ben and Mei stand in a line. Use three name cards (or write A, B, M). How many different orders can you find?	6 orders. Listen for: ABM, AMB, BAM, BMA, MAB, MBA. Listing by who is first helps to find them all.
Puzzle 2 (thinking): Now Ben must stand somewhere in front of Mei (not always right in front). Which of your orders are still allowed?	3 orders: ABM, BAM, BMA. Listen for: Cross out every order where Mei comes before Ben.

Choice during the lesson (groups of 8, online)

- Minutes 10 to 40 (explore, mini-lesson, brain break): everyone does the same. The explore slide has a Go further what-if from minute 10, so a pair that has finished its first task has real work at once. Do not say who is ready for it.
- Minute 40: everyone opens Start here page 1 and works in pairs. A line at the top says: “Do page 1, then Q4 and Q5 on page 2. After that you may choose: Think harder, Go further or the Investigation.” You never announce who is on which page. Invite the whole group, never one child: “When you have finished Start here you can choose Think harder, Go further or the Investigation.” Do not split a pair unless both children agree.
- The first-step help for the Investigation is printed on page 2 of Start here, so every child sees it. Pairs work in breakout rooms, so visit EVERY room by minute 47 (about a minute each, camera on, a parent nearby) and ask “how do you know?”, not “is it done?”. At about minute 52 check in again with two or three children, including children on Start here.
- Minute 67: everyone chooses what to do next: finish page 2, Think harder, Go further or the Investigation. The Investigation has its own slot, minutes 67 to 72, and its debrief opens talk time. Everyone is back together for talk time at minute 72. An Investigation that is not finished goes home; volunteers share one thing at the start of next lesson.
- Next lesson everything starts again from the same pages: nobody is on a track.

DO Warm-up game (minutes 5 to 8). Now the game, then the quick questions (minutes 8 to 10): Relay chain. Each answer is passed on and becomes the start of the next question.

SAY *“Relay chain: I ask the first question. The answer is the start of the next question, so listen carefully and pass it on. If someone is stuck, the group can help with a hint, not the answer.”*

DO After the game, ask the quick questions on the warm-up slides. Use thumbs or a whisper to partners before taking answers.

Ask	Expect
What is the value of the digit 7 in 4732?	700
3/8 + 2/8 = ?	5/8
47 ÷ 5 = ?	9 remainder 2
A rectangle is 6 cm by 4 cm. What is its perimeter?	20 cm
2 kg 50 g = __ g	2050 g

10–25 min Explore (in pairs, with objects)

SAY *"Look at the two stations. Which one do you think will be harder? Write your guess, and we will see at the end."*

DO 1. Two stations, 7 minutes each: List, then Measure.

2. Work in pairs. Read the puzzle card aloud.

3. Before you solve it, decide: draw it, list it or make a table?

4. Write the answer and how you started.

SAY *"Talk to your partner while you work. I will come round and ask you questions."*

IF STUCK Stuck: "How could you start? Draw it, list it, or make a table?"

List station: "Start with the smallest digit in the hundreds place. How many numbers does each digit give?"

Measure station: "Draw the three pieces. What if all three were the same? What is extra?"

Ready for more: make a puzzle card for the next pair.

DO Pairs work from minute 10. If a pair finishes the first task, they may take the Go further what-if on the slide (do not say who). At about minute 15 bring the group back together and take the three questions below, one at a time (minutes 15 to 25).

Ask	Expect
How did you start each puzzle?	Open: draw it, list it, make a table.
How many numbers did the List station make? How do you know you have them all?	6 numbers: 2 for each hundreds digit, listed in order.
Measure station: how long is each piece of ribbon?	30 cm, 40 cm and 50 cm (take away the extra 10 + 20 = 30 cm, 90 cm ÷ 3 = 30 cm).

25–35 min Mini-lesson (you teach, they join in)

DO Show the mini-lesson slides. Think aloud as you work. Say what you notice, what you test and how you check. Children copy how you think.

SAY *"Siti is 4th from the front and 6th from the back. I draw the queue: 3 people in front of Siti, Siti, then 5 people behind her. 3 + 1 + 5 = 9. If I add 4 + 6 I count Siti twice, so I take 1 away: 10 − 1 = 9."*

"A logic grid: Ben does not like curry puff or popiah, so I put two crosses in Ben's row. Only kaya toast is left, so a tick. A tick means crosses in the rest of that column, so nobody else has kaya toast."

"A bar graph: first I read the scale. The lines go 0, 2, 4, 6, so each space is 2. A bar halfway between 6 and 8 shows 7."

Worked example on the slides	Answer

Draw it: a queue. Siti is 4th from the front and 6th from the back of a queue. How many people are in the queue?	9 people: 3 in front, Siti, 5 behind (4 + 6 − 1 = 9).
Make a table: a logic grid. Ben, Siti and Ravi each like a different snack: curry puff, kaya toast, popiah. Ben does not like curry puff or popiah. Ravi does not like curry puff. Who likes what?	Ben: kaya toast. Ravi: popiah. Siti: curry puff.

SAY *"The key idea today is: how to start a puzzle. Stuck? Draw it, list it in order, or make a table. Then check every clue. Say it with me."*

DO Show the key idea slide at minutes 34 to 35. The "common mistakes" on that slide are for you, not the children. Watch for them during the worksheet:

- Queues: adding the two positions and counting the person twice (3rd from front and 5th from back is 7, not 8).
- Bar graphs: reading every space as 1 instead of checking the scale, or not reading the halfway values (45, 55).
- Logic grids: putting a tick but forgetting to cross out the rest of that row and column; or guessing instead of using a clue.

35–40 min Brain break

DO Freeze dance with maths. When the music stops, the tutor shows a 24-hour time; make the clock hands with your arms.

Water. Toilet break for anyone who needs it. Keep it to five minutes.

DO Story break (optional, 2 minutes): after the movement, show the Story break slide (Raymond Smullyan, USA (born in Far Rockaway, New York), 1919 to 2017) and read the story aloud. Link to today: Today you solve logic puzzles with ticks and crosses, just like working out who is a knight and who is a knave. Then offer the try-it as a question for everyone to think about: "undefined" Anyone may pass, nobody is called on by name and nothing is marked. The answer is in the notes of the slide and in the answer key below. Skip it if you are short of time: click straight on to the worksheet.

40–72 min Worksheet (Start here pages, then your choice, with its own Investigation slot at 67–72)

DO Minute 40: give out the workbook. Read the title of Lesson 10 aloud. Say "Pairs first. Talk before you write."

Minutes 40 to 52: page 1 (Start here) in pairs; where the deck has a short teaching slide, it comes first. Pairs are in breakout rooms, so visit EVERY room by minute 47 (about a minute each; camera on, parent nearby), then check in again at about minute 52. Minutes 52 to 54: check page 1 together. Minutes 54 to 58: short teaching before page 2. Minutes 58 to 65: each child works alone on page 2. Minutes 65 to 67: check page 2 together. Minutes 67 to 72: everyone chooses Think harder, Go further or the Investigation (the Investigation has its own slide, minutes 67 to 72). A

child who finishes early starts on one of them at once, and any child may choose them once page 1 and Q4 and Q5 of page 2 are done.

SAY *"Pairs first. Read each question aloud and talk about it before you write. I am not looking for speed. I am looking for good thinking."*

DO Circulate and ask:

- Ask: "How did you start? Why did you choose that?"
- Ask: "Which clue did you use first? Why that one?"
- Ask: "How do you know you have found ALL of them?"

IF STUCK Hint 1: "Draw it: put people or bars on paper."

Hint 2: "Make a table and cross out what cannot be."

Hint 3: "List it in order, smallest first, so you do not miss one."

DO Anyone who wants more may choose Think harder first, then Go further, or the Investigation (its easy first step is on page 2 of Start here). Invite the whole group: "When you have finished Start here you can choose Think harder, Go further or the Investigation." Never announce who is on which page, and do not split a pair unless both children agree.

Think harder and Go further questions

Question	Answer
Workbook Think harder 1: Ali has three coins: 10¢, 20¢ and 50¢. He picks one coin, or two coins, or all three. List every different amount of money he can make, in order. How many amounts are there?	7 amounts: 10¢, 20¢, 30¢ (10 + 20), 50¢, 60¢ (10 + 50), 70¢ (20 + 50), 80¢ (all three). Listing by the number of coins (one, two, three) makes sure none is missed.
Workbook Think harder 2: Ali, Ben and Siti each have a different pet: a cat, a dog and a fish. Ali's pet has no fur. Ben does not have the dog. Who has which pet?	Ali: fish. Ben: cat. Siti: dog.
Workbook Think harder 3: Triangles are made in a row with sticks: 1 triangle uses 3 sticks, 2 triangles use 5 sticks, 3 triangles use 7 sticks. How many sticks are needed for 12 triangles?	25 sticks (add 2 each time: 3 + 11 × 2 = 25)
Workbook Think harder 4: Use the digits 3, 0, 6 and 9 once each. Make the largest even 4-digit number and the smallest odd 4-digit number.	Largest even: 9630. Smallest odd: 3069.

Workbook Go further 5: Make up your own logic puzzle with 3 people and 3 things (colours, pets or sports). Write 3 or 4 clues. Test it on a partner: does it have exactly ONE answer? Is there a clue you could take away and still have only one answer?	Answers vary. Check that the clues fit one arrangement only (try every arrangement of the 3 things). Example: Sam, Lee and Mei wear red, blue and green. Clues: Sam is not wearing red. Lee is not wearing blue. Mei is not wearing green. Sam is not wearing blue. The only answer is Sam green, Lee red, Mei blue. The clue "Mei is not wearing green" could be taken away and there is still only one answer.
Workbook Go further 6: In a queue, Siti is 6th from the front and Ravi is 4th from the back. There are 3 children between them, and Siti is in front of Ravi. How many children are in the queue?	13 (Siti 6th, Ravi 6 + 3 + 1 = 10th from the front, then 3 more behind Ravi)
Workbook Go further 7: A bar graph shows the paper collected by 4 classes. 3P1 collected 10 kg more than 3P2. 3P3 collected the same as 3P1. 3P4 collected 5 kg less than 3P2. Altogether they collected 135 kg. How much did each class collect?	3P1 40 kg, 3P2 30 kg, 3P3 40 kg, 3P4 25 kg (take away the extras 10 + 10 − 5 = 15: 120 kg ÷ 4 = 30 kg for 3P2)

These are the questions on the Think harder and Go further pages in the workbook. More are in the separate Question Bank.

72–80 min Talk time (Investigation debrief first)

DO Bring the group back together (camera on). First 3 minutes (72 to 75): the Investigation debrief. Ask for volunteers to share what they noticed, then show the model answer (in the Investigation row at the end of this lesson's answer key and in the notes of the Investigation slide). An unfinished Investigation can be shared at the start of next lesson. Then the talk question. Ask for volunteers first. A pass is always fine, and so is sharing with a partner. Do not call a child by name cold, and keep no public tallies or rankings. Nobody is called on by name. Come back to sneak-peek puzzle 2: "Now we know the idea, how could it help?"

SAY *"Which start helped you most today: draw it, list it or make a table? Why?"*

"Why do we take away 1 in a queue puzzle?"

Discussion question	Listen for
Choose one puzzle. How did you START it, and how do you know your answer is right? (Share, share with a partner, or pass.)	Open answers for the first. Queue: the person is counted from the front AND from the back, so they are counted twice; take away 1.

80–90 min Exit ticket (80–86), look back (86–88) and closing (88–90)

DO Minutes 80 to 86. Give each child a slip of paper (or ask them to write on their journal page). No talking, no helping. Show the exit ticket slide. Children show the answer to you on camera in the open call (or keep it to share if they want); no photos and no private messages. Never mark or read answers out in front of the group.

Exit ticket	Answer
Ali is 4th from the front and 4th from the back of a queue. How many people are in the queue? Which start did you use?	7 people (3 in front, Ali, 3 behind: 4 + 4 − 1 = 7). Draw it.

Look back (minutes 86 to 88, two minutes)

SAY *"Turn to the last page of today's lesson and find Look back. Tick or write one thing for each line: something you did well, the hardest part, and what helped you. There are no right answers, and you may pass on any line."*

"Would anyone like to share something they did well, or what helped? Thank you. When you got stuck and tried something else, that is exactly what mathematicians do."

DO Give a full 2 minutes. Never collect the page, mark it or read it out in front of the group: children keep it. You cannot see the paper. Tell the WHOLE group: "If you ticked still stuck, say check-in please in the open class, or ask your grown-up to tell me." Then talk it through with that child in the open session, with a parent nearby. No hand signals that single a child out, never a private message. Open the next lesson with the whole group: "What is still tricky from last time?" Invite volunteers only. Praise the strategy ("you drew it and that got you unstuck"), not the speed or the answer.

SAY *"You are real inquirers now: you notice, test, and explain why. Here is your certificate."*

"Give each child a certificate and tell them one specific thing they did well in the program."

DO Minutes 88 to 90. Stamp the inquiry journal when a child has shown you their exit ticket. Mark S, D or H in the progress tracker at the back of this guide (it is for you only). If a child is marked D or H, plan two minutes of quiet catch-up at the start of the next lesson.

Extension for children who want more

Write your own logic grid puzzle with 3 people and 3 things. Test it on a friend: does it have exactly one answer? Which clue could you take away?

Answer key (workbook)

Q1 (a) Kopi (b) 25 cups (55 − 30) (c) 50 cups (200 − 45 − 30 − 55 − 20); the bar reaches the 50 line (d) Milo and Kopi (45 + 55 = 100)

Q2 Ali: football. Mei: chess. Ravi: swimming. Siti: choir. (Clue 2: Mei and Ravi are not in the choir; with clue 1, Siti is. Clue 3: the swimmer is Ravi or Siti, so Ravi. Ali is not chess, so football; Mei chess.)

Q3 (a) 7 people (3 + 5 − 1) (b) 6 people (Ali 2nd, 3 between, Siti 6th and last)

Q4 (a) 258, 285, 528, 582, 825, 852 (b) 2 odd numbers: 285 and 825 (c) 582 (18 away from 600)

Q5 Table: 13, 16, 19 sticks (add 3 each time) (a) 25 sticks (b) 10 squares

Q6 411, 321, 213 (3 numbers: the ones digit is 1, 3 or 5; list each case)

Sneak-peek puzzles Puzzle 1: 6 orders Puzzle 2: 3 orders: ABM, BAM, BMA

Investigation (Handshakes: draw it, list it, make a table) Model answer and what to look for: 2 friends: 1; 3: 3; 4: 6; 5: 10; 6: 15. The extra amount goes up by 2, 3, 4, 5, so 7 friends: 21, 8: 28, 9: 36, 10: 45. Reason: each new friend shakes hands with everyone already there. 10 friends: 9 + 8 + 7 + … + 1 = 45. 20 handshakes is impossible: 6 friends give 15 and 7 friends give 21 (checked by computer for every group size). Look for: a drawing or an organised list, the pattern in the jumps, and a sensible check of the 20 claim using the table. Extension: how many for 20 friends? (190)

Story break try-it (Raymond Smullyan, optional) Ana is a knight and Ben is a knave. If Ben were a knight, both would be knights, but then Ana's words 'Ben is a knave' would be a lie, which a knight never tells. So Ben is a knave, his words are false, and Ana's words are true, so she is a knight.

Go further (explore slide) A 120 cm ribbon is cut into 4 pieces, each 10 cm longer than the last. How long is each? Answer: 15 cm, 25 cm, 35 cm and 45 cm (take away 10 + 20 + 30 = 60, 60 cm left, 60 / 4 = 15).

Level 3 stories: mathematicians

Each lesson has an optional two-minute Story break about a real mathematician, at the brain break (after the mini-lesson, before the worksheet). The story is on its own slide, the child workbook has a small Story corner with the same try-it question (no answer on the child page), and the answer is in the answer key for the lesson. Skip it whenever you are short of time. Nobody is asked to answer by name, and nothing is marked. Stories marked as a legend are said to be only a story, and the slides say so.

Lesson	Person	Country	Years	Link to the lesson
1	Rozsa Peter	Hungary (Budapest)	1905 to 1977	Today you continue number patterns, and a rule that makes each number from the one before is the idea of recursion that she studied.
2	Joan Clarke	England (London, Cambridge, Bletchley Park)	1917 to 1996	Today you crack letter puzzles by trying possibilities and working out what must be true, and codebreakers also cut down possibilities to find the secret.
3	John Napier	Scotland	1550 to 1617	Napier's rods are made of times tables, so they turn multiplication into reading a table and adding, just like the tables of 6, 7, 8 and 9 we explore today.
4	Bhaskara II	India	1114 to 1185	Today we solve word problems by drawing a bar model, and Bhaskara's book is full of stories that ask us to find an unknown number.
5	Sun Zi	China	lived more than 1,500 years ago (exact dates unknown)	Today's "what number am I?" riddles use division with remainders, just like Sun Zi's riddle.
6	Abu Bakr al-Hassar	Morocco / the Islamic West (sources differ between the Maghreb and Spain)	12th century (exact dates unknown)	Today you read and compare fractions, and the bar between the top and bottom numbers is the one al-Hassar described.

7	Karen Uhlenbeck	USA (born in Cleveland, Ohio)	born 1942	Today you find area and perimeter, and she studied shapes that use the smallest area for the edge that holds them.
8	Ibn al-Haytham	Born probably in Basra (now Iraq); worked mostly in Cairo, Egypt	about 965 to about 1040	Today you look for angles and perpendicular lines, and light bouncing off a mirror makes equal angles on either side of the perpendicular.
9	Zu Chongzhi	China (born in Jiankang, now Nanjing)	429 to about 500 (sources differ)	Today you work with units of time, and Zu measured the year in days, hours and parts of a day.
10	Raymond Smullyan	USA (born in Far Rockaway, New York)	1919 to 2017	Today you solve logic puzzles with ticks and crosses, just like working out who is a knight and who is a knave.

Notes on accuracy

- Lesson 2 (Joan Clarke): Sources differ slightly on the year she became deputy head and on her honours, so the story gives no years for those. Letter puzzles are our link; the sources do not say she did school-style puzzles.
- Lesson 4 (Bhaskara II): A story often told says the Lilavati was named after Bhaskara's daughter (historians have no proof; it is a legend). The necklace problem in try-it is our own, written in the style of the book, not a copy of his.
- Lesson 5 (Sun Zi): Dates and author of the Sunzi Suanjing are uncertain; the story says so.
- Lesson 6 (Abu Bakr al-Hassar): His dates and place are uncertain, and the story says so. The claim that Fibonacci's use of the bar came after al-Hassar is stated by MAA and Muslim Heritage. The sources do not say whether Fibonacci learned the bar from al-Hassar.
- Lesson 7 (Karen Uhlenbeck): The story uses the soap film only as a picture of 'least area'; the sources say a soap film settles into a minimal surface.
- Lesson 8 (Ibn al-Haytham): The house-arrest story is a legend (MacTutor says it 'appears' true; Wikipedia calls it a legend). Birth place and death place are uncertain, and sources give 1039 or 1040. The equal-angles rule is the modern law of reflection and is not claimed as his own words.
- Lesson 9 (Zu Chongzhi): Wikipedia and MacTutor disagree on his dates and on the polygon he used, so the story gives neither polygon nor an exact death year. The 'within about a minute' comparison is our own check: 365.24281481 - 365.24219878 days is about 53 seconds, and MacTutor says about 50 seconds. Only MacTutor says the calendar was never put into use, so the story leaves it out.

Progress tracker

After each exit ticket, write S (secure: correct and can explain), D (developing: correct but cannot explain yet) or H (needs help). Use the notes for what to revisit next lesson.

Student	L1	L2	L3	L4	L5	L6	L7	L8	L9	L10	Notes

Ending the program

- In Lesson 10, give each child a certificate and their completed inquiry journal.
- Send parents a short message through the approved channel: one thing their child did well, and one thing to keep practising.
- Delete or hand over all Student Data within 7 days after the last lesson (this is in the tutor agreement).

Tutor checklist

Before each lesson

- Read the lesson plan and solve the worksheet yourself without looking at the key.
- Check materials are ready, or the online set-up has been sent.
- Test the platform, screen share and annotation tools 10 minutes early (online).
- Print or open the slide deck for the lesson.

During the lesson

- Circulate and ask questions rather than giving answers.
- Use only the approved channels to contact parents. Never contact a child outside the lesson platform. No private messages to children.
- Do not take photos or recordings unless the Company has approved it and parents have consented.
- If you have a concern about a child's safety or welfare, tell the Company straight away.

After the lesson

- Stamp inquiry journals and fill in the tracker.
- Keep answer keys secure. Collect worksheets face to face only if a child offers them; the look-back page is never collected.
- Note any child who needs help, and plan a short catch-up at the start of the next lesson.

Math Inquirers Holiday Program

Notice it. Test it. Explain why.

A 10-lesson holiday maths program for children whose school year covers Primary 3 topics

Your child will spend the holidays as a young mathematician: noticing patterns, testing ideas with real objects and explaining their thinking. Lessons are live online in small groups, hands-on, and built around thinking, not speed.

What your child will do

- Lesson 1, Number Sleuths: numbers to 10 000
- Lesson 2, Cryptarithm Crackers: addition and subtraction
- Lesson 3, Times Table Treasure: tables of 6, 7, 8 and 9
- Lesson 4, Model Makers: word problems: bar models
- Lesson 5, Remainder Riddles: division
- Lesson 6, Fraction Walls: fractions
- Lesson 7, Area Architects: area and perimeter
- Lesson 8, Angle and Line Detectives: angles, perpendicular and parallel lines
- Lesson 9, Measure and Money Masters: length, mass, volume, money and time
- Lesson 10, Challenge Day: mixed puzzles: bar graphs and logic

How you can help

- Ask “How do you know?” when your child tells you an answer.
- Let your child try first. Please sit nearby, but let them find the answer.
- Print the workbook and cut-outs, and keep scissors, a ruler and a pencil ready.

Practical details

Dates	[start date] to [end date], two lessons a week
Time	[day] and [day], [start time] to [end time]
Where	Live online video class (link sent after registration)
Group size	[number] children
Fee	S$500 introductory (regular S$650), workbook included, payable in 3 instalments
Contact	[name, phone or email]

Math Inquirers Holiday Program is an independent enrichment program. It is inspired by enrichment-style thinking skills but is not affiliated with, run by or endorsed by the Ministry of Education or any school.